Д/З 6.1.1 - 6.1.24

Д/З

6.1.3

$$f(x) = \frac{x^2+4}{x^3+1}$$

□ $D(f): x^3+1 \neq 0$

$x^3 \neq -1$

$x \neq -1 \Rightarrow D(f) = (-\infty; 1) \cup (1; \infty)$ ■

6.1.4

$$f(x) = \sin\frac{1}{|x|-2}$$

□ $D_1(\sin x) = (-\infty; \infty)$ $D_2\left(\frac{1}{|x|-2}\right)$: $|x|-2 \neq 0$

$|x| \neq 2$

$D(|x|) = (-\infty; \infty)$

$x \neq \pm 2$

$D(f) = (-\infty; -2) \cup (-2; 2) \cup (2; \infty)$

■

6.1.5

$f(x) = \log_3(-x)$

□ $D(f): \ -x > 0$

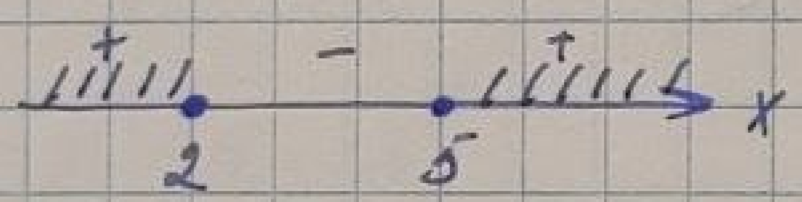
$x < 0 \Rightarrow D(f) = (-\infty; 0)$ ■

6.1.6

$f(x) = \sqrt[4]{x^2 - 7x + 10}$

□ $D(f): \ x^2 - 7x + 10 \geq 0$

$(x-2)(x-5) \geq 0$



$\Rightarrow D(f) = (-\infty; 2] \cup [5; \infty)$ ■

6.1.7

□ $f(x) = x^2 + tg\,x$

$D_1(x^2) = (-\infty; \infty)$

$D_2(tg\,x) \in \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\}, k \in \mathbb{Z}$

$D(f) = D_1 \cap D_2 = \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\}, k \in \mathbb{Z}$ ■

6.1.8

$f(x) = \sqrt{x-7} + \sqrt{10-x}$

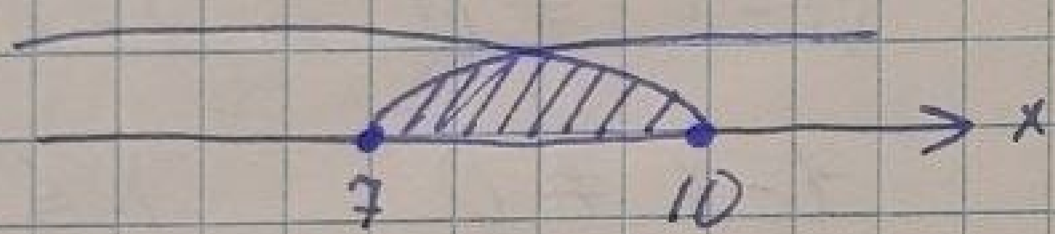
$D_1(\sqrt{x-7}) : x - 7 \geq 0$

$x \geq 7 \Rightarrow D_1(\sqrt{x-7}) = [7; \infty)$

$D_2(\sqrt{10-x}) : 10 - x \geq 0$

$x \leq 10 \Rightarrow D_2 = (-\infty; 10]$

$D(f) = D_1 \cap D_2 = [7; 10]$



6.1.9

$$f(x) = \frac{\ln x}{\sqrt{|x^2-2|}}$$

$D_1(\ln x) = x > 0 \Rightarrow D_1 = (0; \infty)$

$D_2(\sqrt{|x^2-2|}) : |x^2-2| \geq 0$

$x \in \mathbb{R} \Rightarrow D_2 = (-\infty; \infty)$

$D_3\left(\frac{1}{\sqrt{|x^2-2|}}\right) : \sqrt{|x^2-2|} \neq 0$

$|x^2-2| \neq 0$

$x^2 - 2 \neq 0$

$x^2 \neq 2$

$x \neq \pm\sqrt{2} \Rightarrow D_3 = (-\infty; -\sqrt{2}) \cup (-\sqrt{2}; \sqrt{2}) \cup$

$\cup (\sqrt{2}; \infty)$

$D(f) = D_1 \cap D_3 = (0; \sqrt{2}) \cup (\sqrt{2}; \infty)$

■

6.1.10

$f(x) = \sqrt[4]{x+2} + \dfrac{1}{\sqrt[6]{1-x}}$

□ $D_1(\sqrt[4]{x+2})$: $x+2 \geq 0$

$x \geq -2 \Rightarrow D_1 = [-2; \infty)$

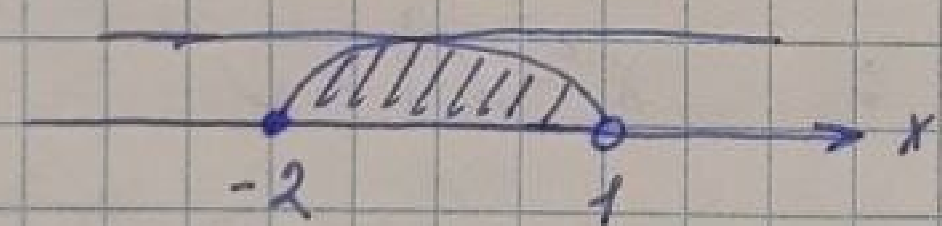
$D_2\left(\dfrac{1}{\sqrt[6]{1-x}}\right)$: $\sqrt[6]{1-x} \neq 0$ $\quad 1-x \geq 0$

$1-x \neq 0$ $\quad x \leq 1$

$x \neq 1$

$\Rightarrow D_2 = (-\infty; 1)$

$D(f) = D_1 \cap D_2 = [-2; 1)$



■

6.1.12

$f(x) = \arccos(x-2) - \ln(x-2)$

□ $D(\arccos x) = [-1; 1]$

$-1 \leq x-2 \leq 1 \quad | +2$

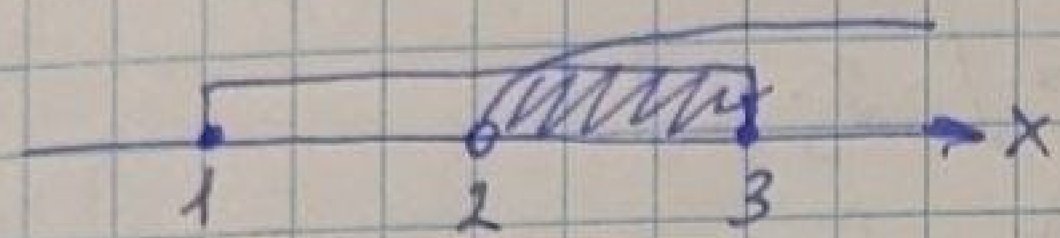
$1 \leq x \leq 3$

$\Rightarrow D_1(\arccos(x-2)) = [1; 3]$

$D_2\,(\ln(x-2))$: $x-2>0$

$x>2 \Rightarrow (2;\infty)$

$D(f) = D_1 \cap D_2 = (2;3]$



■

6.1.14

$f(x) = x^2 - 8x + 20$

□ $f(x) = x^2 - 8x + 20 = (x^2 - 8x + 16) - 16 + 20 =$

$= (x-4)^2 + 4$

$(x-4)^2 + 4 \geq 0$

$(x-4)^2 \geq -4 \Rightarrow [-4;\infty)$

$\Rightarrow E(f) = [-4;\infty)$ ■

6.1.15

$f(x) = 3^{-x^2}$

□ $-x^2$: 2 - чет степ $\Rightarrow x^2 \geq 0$

$3^{-x^2} = \dfrac{1}{3^{x^2}}$; $E(3^{x^2}) = [1;\infty)$

$E\left(\dfrac{1}{3^{x^2}}\right) = \left[\dfrac{1}{1};\dfrac{1}{\infty}\right) = [1;0)$

$\Rightarrow E(3^{-x^2}) = (0;1]$ ■

6.1.16

$f(x) = 2\sin x - 7$

$\square$ $E(\sin x) = [-1; 1]$

$E(2\sin x) = [-2; 2]$

$E(2\sin x - 7) = [-2-7; 2-7] =$

$= [-9; -5]$ $\blacksquare$

6.1.17

$f(x) = \frac{1}{x} + 4$

$\square$ $E(\frac{1}{x}) = (-\infty; 0) \cup (0; \infty)$

$E(\frac{1}{x} + 4) = (-\infty; 0+4) \cup (0+4; \infty) =$

$= (-\infty; 4) \cup (4; \infty)$ $\blacksquare$

6.1.20

$f(x) = \frac{x+3}{x^2-1}$

$\square$ 1) $f(0) = \frac{0+3}{0^2-1} = \frac{3}{-1} = -3$

2) $f(-2) = \frac{-2+3}{(-2)^2-1} = \frac{1}{4-1} = \frac{1}{3}$

3) $f(\sqrt{2}) = \frac{\sqrt{2}+3}{(\sqrt{2})^2-1} = \frac{\sqrt{2}+3}{2-1} = \sqrt{2}+3$

5)

6

$\square$

4) $f(-x) = \dfrac{-x+3}{(-x)^2-1} = \dfrac{-x+3}{x^2-1}$

5) $f\left(\frac{1}{x}\right) = \dfrac{\frac{1}{x}+3}{\left(\frac{1}{x}\right)^2-1} = \dfrac{3+\frac{1}{x}}{\frac{1}{x^2}-1}$

6) $f(a+1) = \dfrac{a+1+3}{(a+1)^2-1} = \dfrac{a+4}{a^2+2a+1-1} =$

$= \dfrac{a+4}{a^2+2a}$

7) $f(a)+1 = \dfrac{a+3}{a^2-1}+1 = \dfrac{a+3+a^2-1}{a^2-1} =$

$= \dfrac{a^2+a+2}{a^2-1}$

8) $f(2x) = \dfrac{2x+3}{(2x)^2-1} = \dfrac{2x+3}{4x^2-1}$ ■

6.1.21

$f(x) = x^3 \cdot 2^x$

□ 1) $f(1) = 1^3 \cdot 2^1 = 1 \cdot 2 = 2$

2) $f(-3) = (-3)^3 \cdot 2^{-3} = -27 \cdot \frac{1}{8} = -\frac{27}{8}$

3) $f(-\sqrt[3]{5}) = (-\sqrt[3]{5})^3 \cdot 2^{-\sqrt[3]{5}} = -5 \cdot \dfrac{1}{2^{\sqrt[3]{5}}} =$

$= -\dfrac{5}{2^{\sqrt[3]{5}}}$

4) $f(-x) = (-x)^3 \cdot 2^{-x} = -x^3 \cdot \dfrac{1}{2^x} = -\dfrac{x^3}{2^x}$

4) $\varphi(z+3) = \dfrac{\sqrt{z+3+5}}{(z+3)^2} = \dfrac{\sqrt{z+8}}{z^2+6z+9}$

5) $\varphi(2t-1) = \dfrac{\sqrt{2t-1+5}}{(2t-1)^2} = \dfrac{\sqrt{2t+4}}{4t^2-4t+1}$

■

6.1.24

1) $f(x) = \dfrac{\sin x}{x}$

□ $D(f):\ x \neq 0 \Rightarrow D(f) = (-\infty; 0) \cup (0; \infty)$

$f(-x) = \dfrac{\sin(-x)}{-x} = \dfrac{-\sin x}{-x} = \dfrac{\sin x}{x} = f(x)$ ■

$\Rightarrow f(x)$ - чет

2) $f(x) = x^5 + 3x^3 - x$

□ $D(f) = (-\infty; \infty)$

$f(-x) = (-x)^5 + 3(-x)^3 - (-x) = -x^5 - 3x^3 + x =$

$= -(x^5 + 3x^3 - x) = -f(x)$

$\Rightarrow f(x)$ - нечет. ■

3) $f(x) = \sqrt{x}$

□ $D(f):\ x \geq 0 \Rightarrow D(f) = [0; \infty)$

$f(-x) = \sqrt{-x} \neq \pm f(x) \Rightarrow$

$\Rightarrow f(x)$ - общ. вид ■

4) $\varphi(z+3) = \dfrac{\sqrt{z+3+5}}{(z+3)^2} = \dfrac{\sqrt{z+8}}{z^2+6z+9}$

5) $\varphi(2t-1) = \dfrac{\sqrt{2t-1+5}}{(2t-1)^2} = \dfrac{\sqrt{2t+4}}{4t^2-4t+1}$

$\blacksquare$

6.1.24

1) $f(x) = \dfrac{\sin x}{x}$

$\square$ $D(f): x \neq 0 \Rightarrow D(f) = (-\infty; 0) \cup (0; \infty)$

$f(-x) = \dfrac{\sin(-x)}{-x} = \dfrac{-\sin x}{-x} = \dfrac{\sin x}{x} = f(x)$

$\Rightarrow f(x)$ - чет $\blacksquare$

2) $f(x) = x^5 + 3x^3 - x$

$\square$ $D(f) = (-\infty; \infty)$

$f(-x) = (-x)^5 + 3(-x)^3 - (-x) = -x^5 - 3x^3 + x =$

$= -(x^5 + 3x^3 - x) = -f(x)$

$\Rightarrow f(x)$ - нечет. $\blacksquare$

3) $f(x) = \sqrt{x}$

$\square$ $D(f): x \geq 0 \Rightarrow D(f) = [0; \infty)$

$f(-x) = \sqrt{-x} \neq \pm f(x) \Rightarrow$

$\Rightarrow f(x)$ - общ. вид $\blacksquare$

4) $f(x) = \arcsin x$

$\square$ $D(f) = [-1; 1]$

$f(-x) = \arcsin(-x) = -\arcsin x = -f(x) \Rightarrow$

$\Rightarrow f(x)$ - нечет

$\blacksquare$

5) $f(x) = \sin x + \cos x$

$\square$ $D_1(\sin x) = (-\infty; \infty)$

$D_2(\cos x) = (-\infty; \infty)$

$D(f) = D_1 \cap D_2 = (-\infty; \infty)$

$f(-x) = \sin(-x) + \cos(-x) =$

$= -\sin x - \cos x = -(\sin x + \cos x) =$

$= -f(x) \Rightarrow f(x)$ - нечет

$\blacksquare$

6) $f(x) = |x| - 2$

$\square$ $D_1(|x|) = (-\infty; \infty)$

$D(f) = (-\infty; \infty)$

$f(-x) = |-x| - 2 = |x| - 2 = f(x) \Rightarrow$

$\Rightarrow f(x)$ - чет.

$\blacksquare$

7) $f(x) = \frac{3}{x^2 - 1}$

$\square$ $D(f): x^2 - 1 \neq 0$

$x^2 \neq 1$

$x \neq \pm 1 \Rightarrow D(f) = (-\infty; -1) \cup$

$\cup (-1; 1) \cup (1; \infty)$

$f(-x) = \frac{3}{(-x)^2 - 1} = \frac{3}{x^2 - 1} = f(x) \Rightarrow$

$\Rightarrow f(x)$ - чет

$\blacksquare$

8) $f(x) = x \cdot e^x$

$\square$ $D(e^x) = (-\infty; \infty)$

$D(f) = (-\infty; \infty)$

$f(-x) = -x \cdot e^{-x} = -\frac{x}{e^x} \neq \pm f(x) \Rightarrow$

$\Rightarrow f(x)$ - общ. вид.

$\blacksquare$

Д/З 6.1.25 – 6.1.38

Д/З

6.1.26

1) $f(x) = \cos\frac{x}{4}$

$\cos x: \ T = 2\pi$

тогда

$\cos\frac{x}{4}: \ T = \frac{2\pi}{4} = \frac{\pi}{2}$

2) $f(x) = |x|$

ф-я модуля

при $x > 0$ ф-я $\uparrow$

[при $x < 0$ ф-я $\downarrow$] $\Rightarrow$ ф-я непериод.

на $[0; \infty) \Rightarrow \forall x \in (-\infty, \infty) \ \exists T$

3) $f(x) = tg(2x-1)$

$tg(2x-1) = tg(2x+\pi-1) =$

$= tg\left(2\left(x+\frac{\pi}{2}\right)-1\right)$

$tg(2x-1): \ T = \frac{\pi}{2}$

4) $f(x) = \sin\frac{x}{2} - ctg x$

$\sin\frac{x}{2}: T = \frac{2\pi}{2} = \pi$

$ctg x: \ T = \pi$

$\sin\left(\frac{x}{2}+T_1\right)-ctg\,(x+T_2)$

$НОК\,(T_1, T_2) = НОК\,(\pi;\pi) = \pi$

5) $f(x) = \sin 3x \cdot \cos 3x$

$\sin 3x : T = \frac{2\pi}{3}$

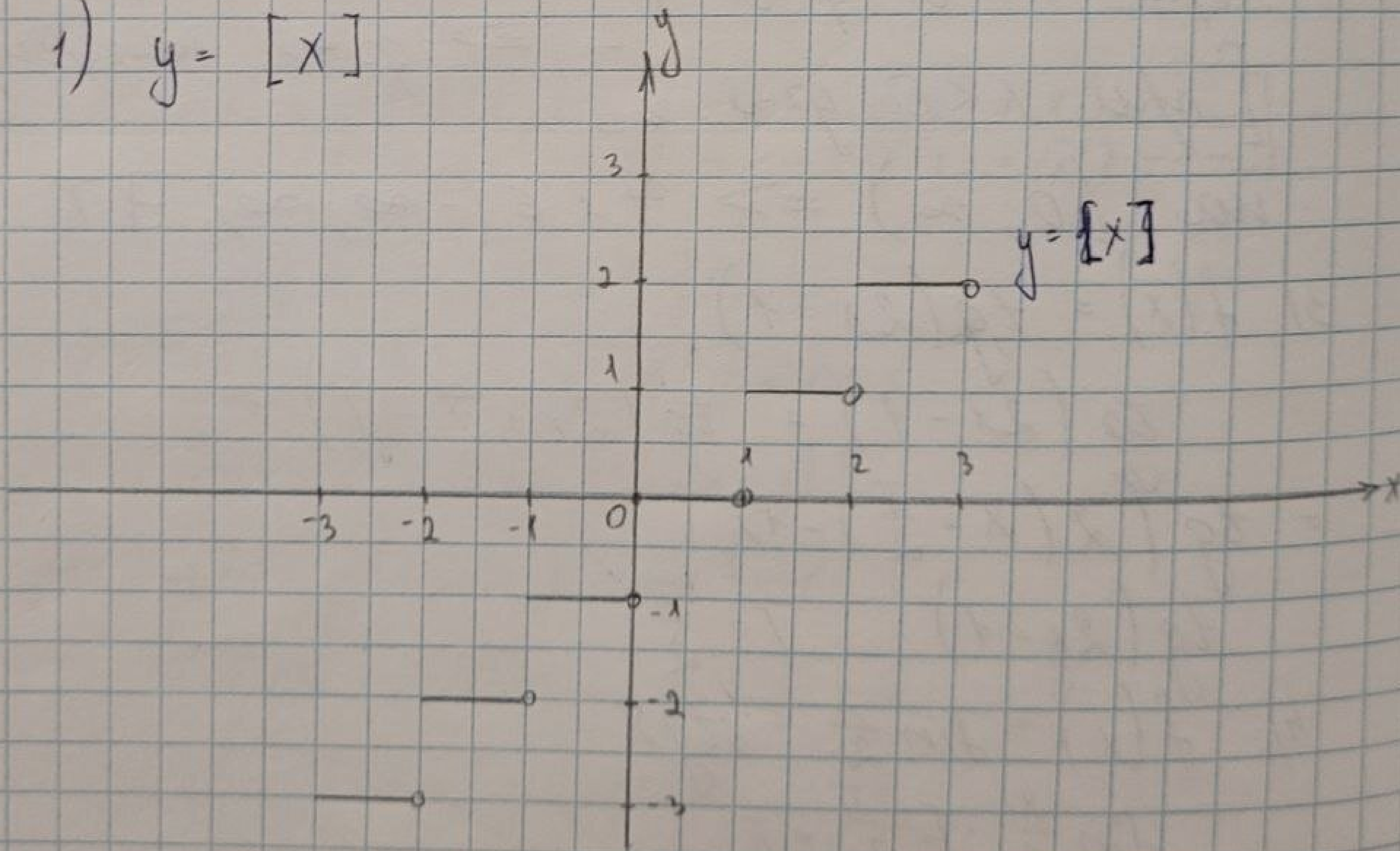
$\cos 3x : T = \frac{2\pi}{3}$

~~sin 3x cos 3x = sin~~

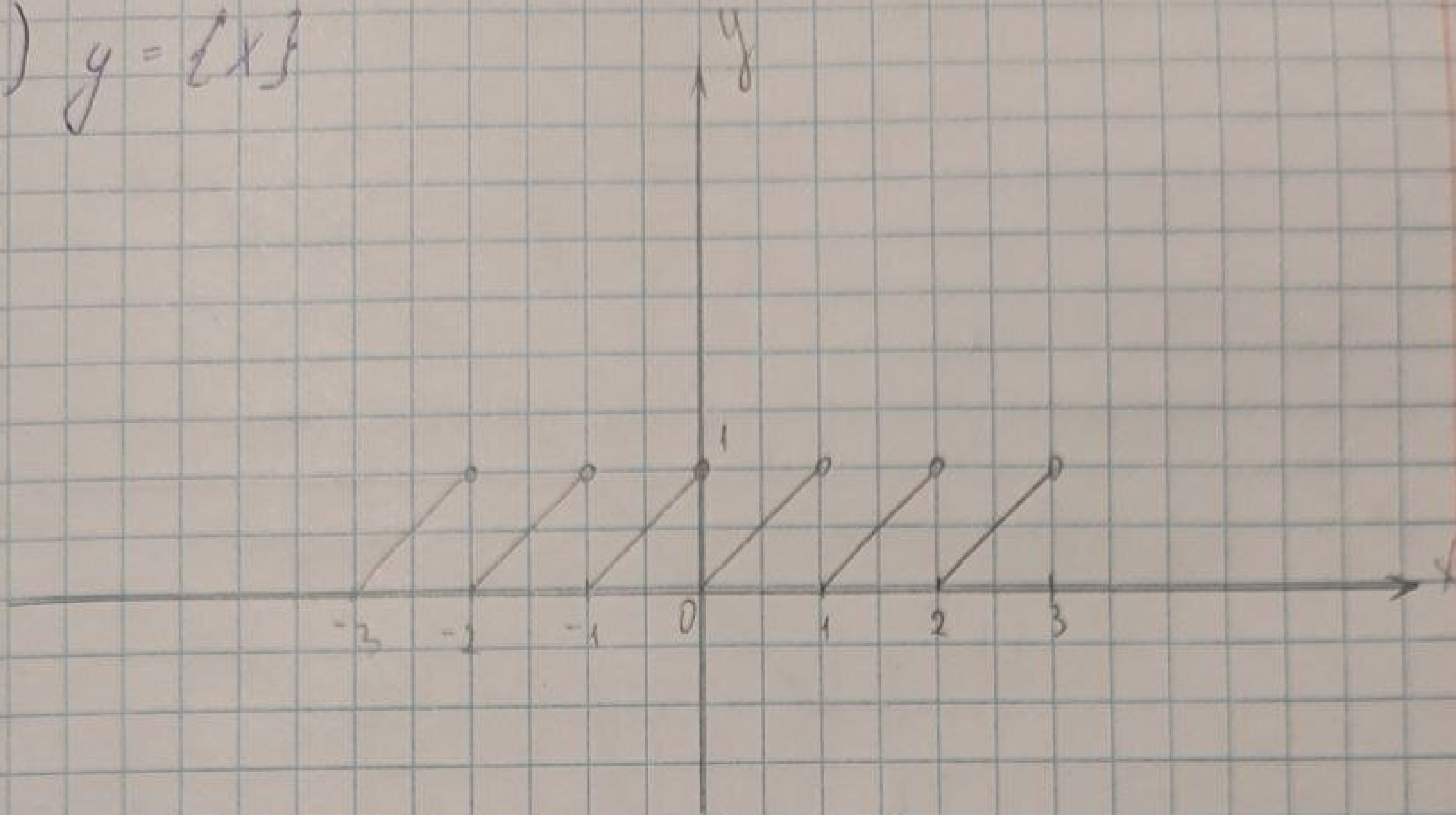
$НОК\,(T_1; T_2) = НОК\left(\frac{2\pi}{3}; \frac{2\pi}{3}\right) = \frac{2\pi}{3}$

6.1.27

1) $y = [x]$



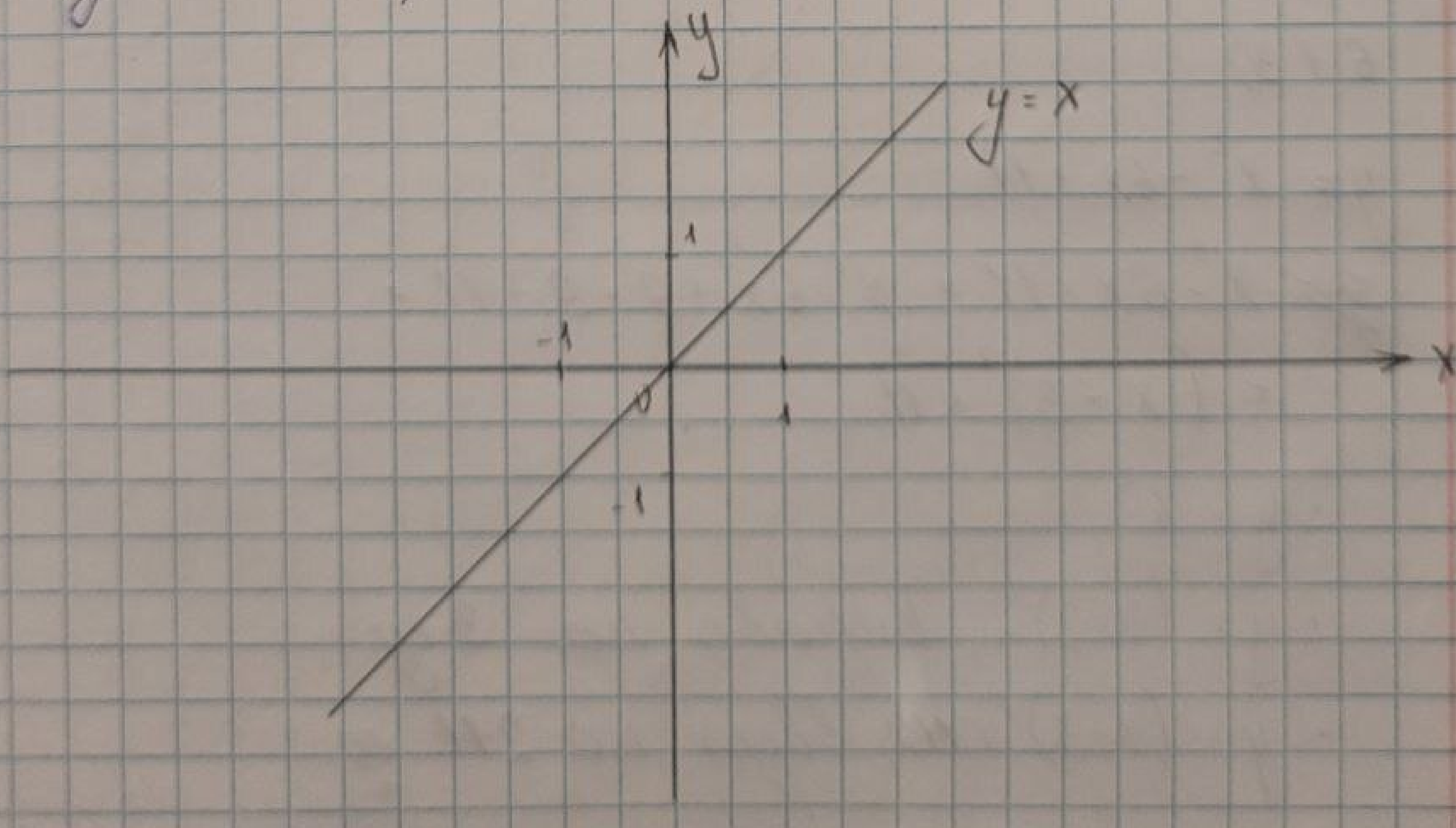
2) $y = \{x\}$

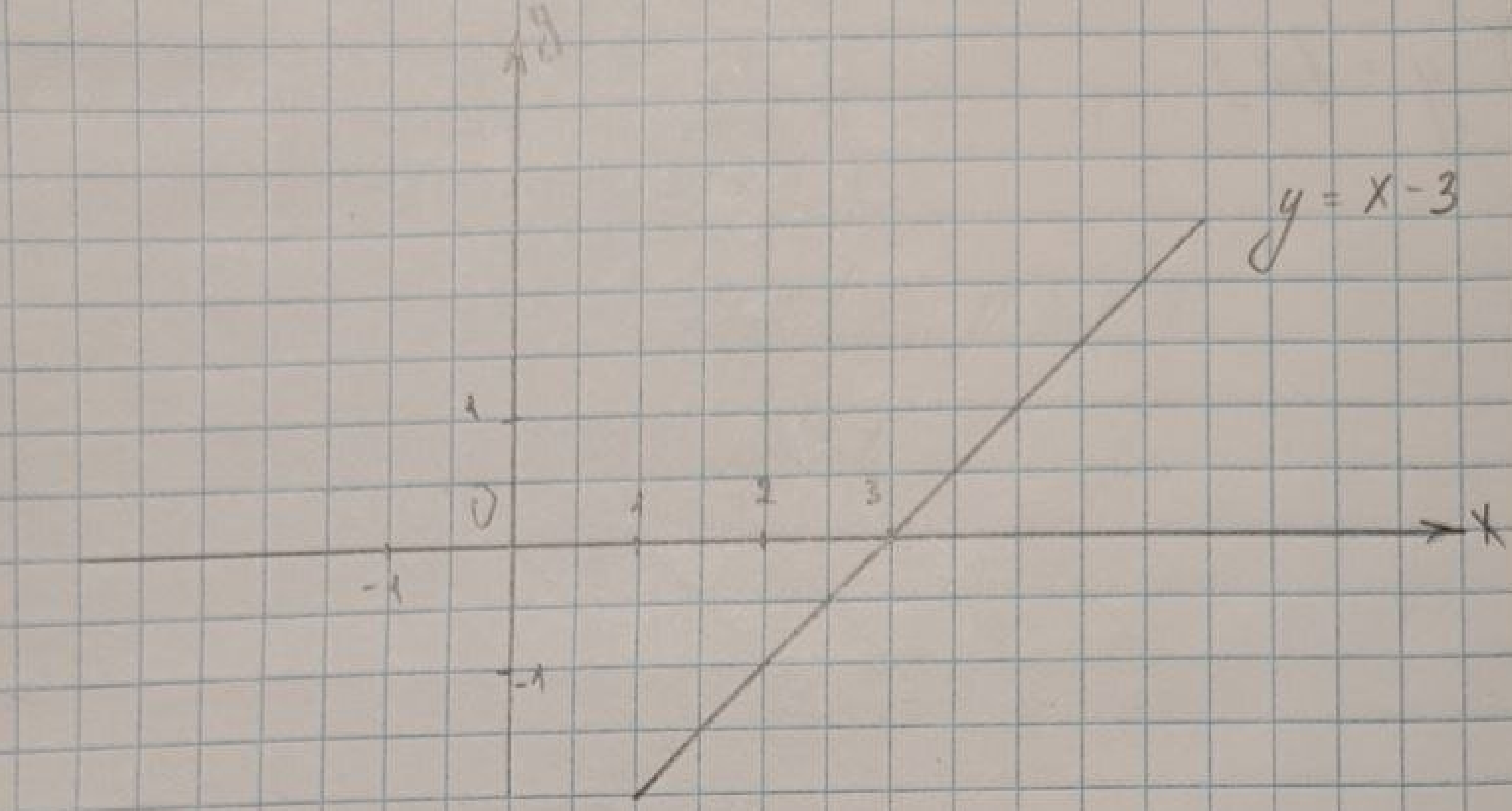


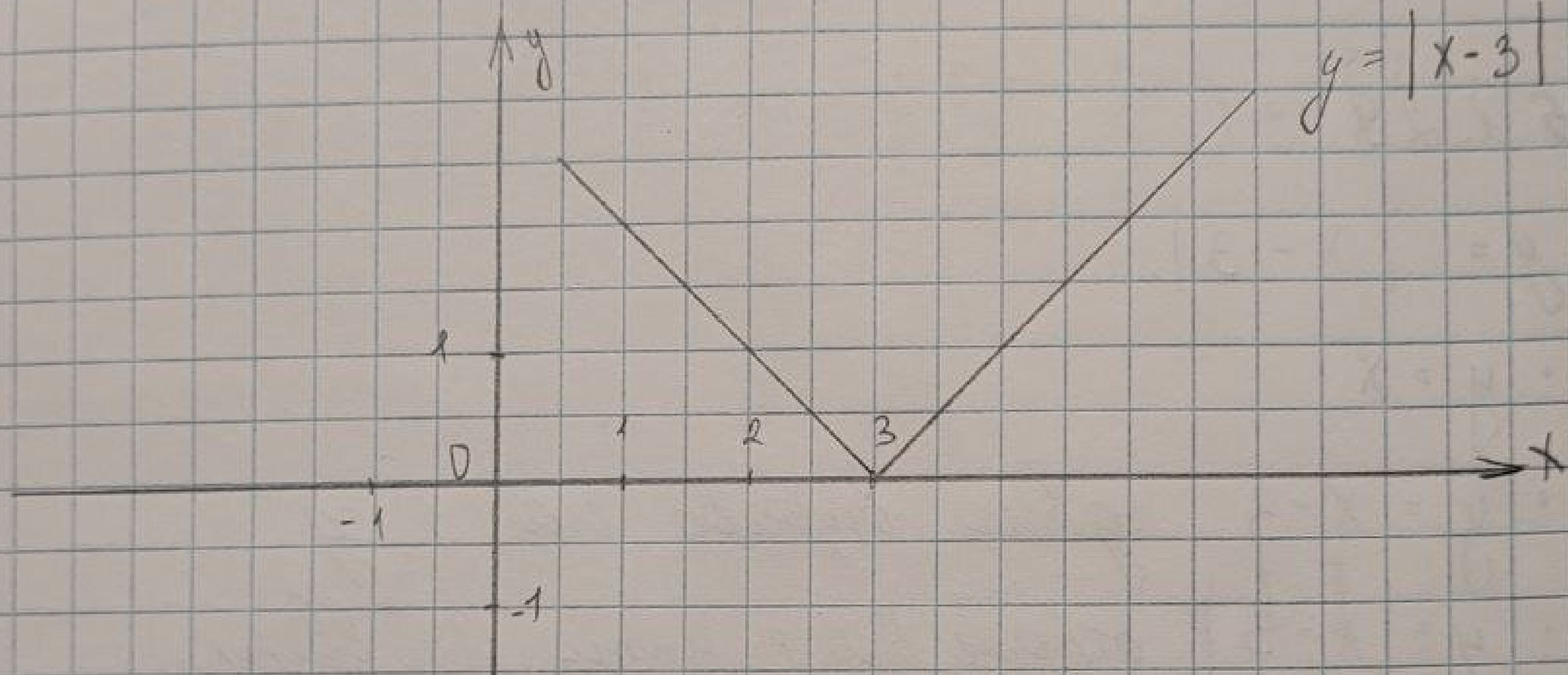
6.1.29

$y = |x - 3|$

- $y = x$
- $y = x - 3$, сдвиг вправо на 3 ед.
- $y = |x - 3|$, отразить все т. ниже Ox вверх.







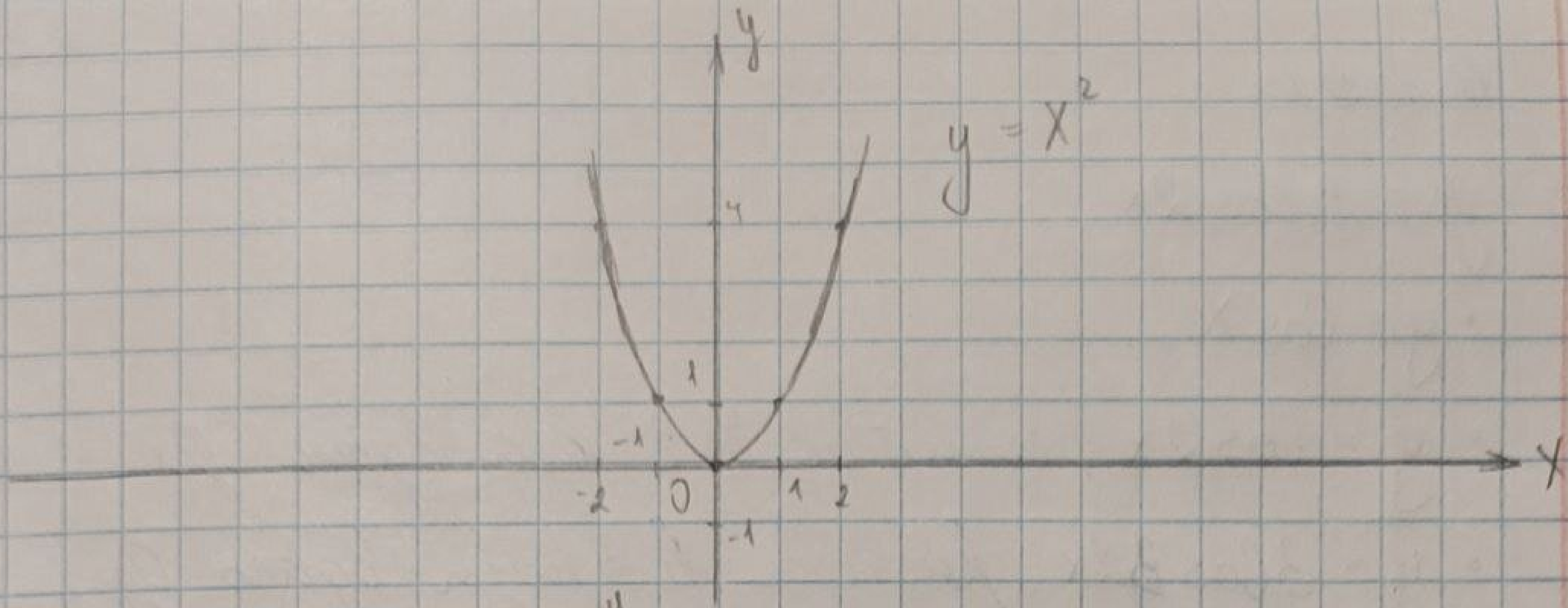
6.1.30

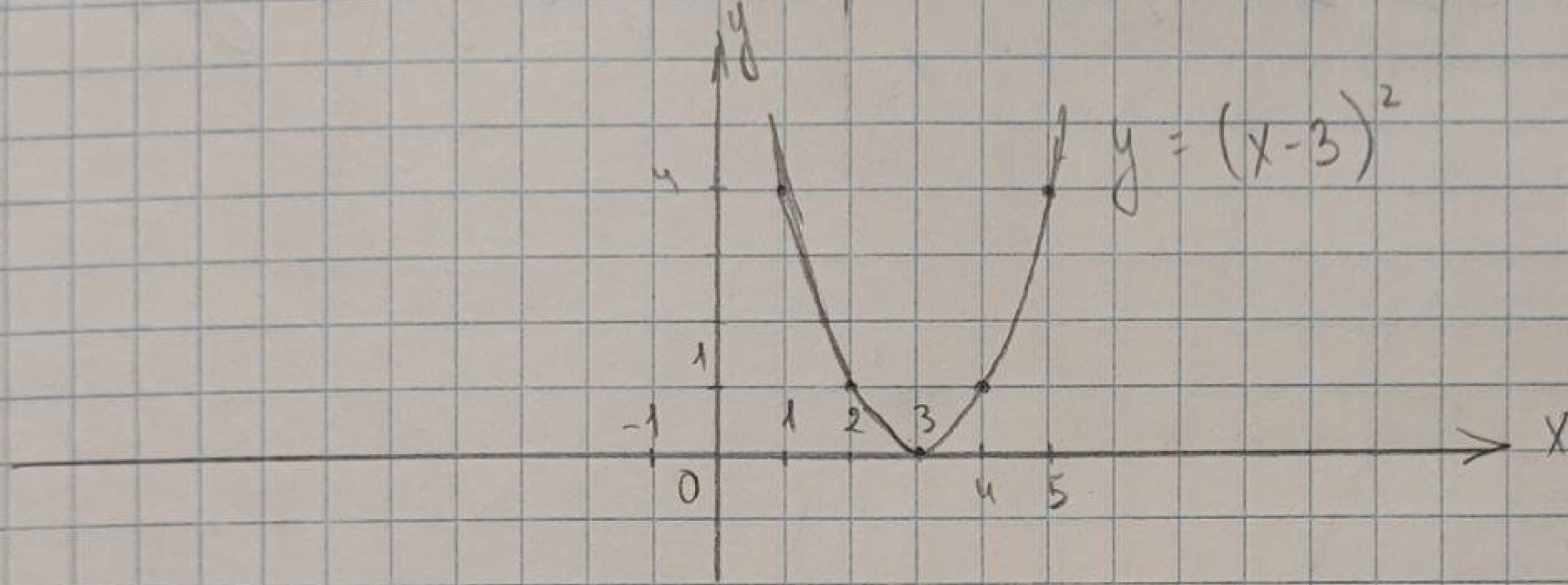
$y = x^2 - 6x + 11$

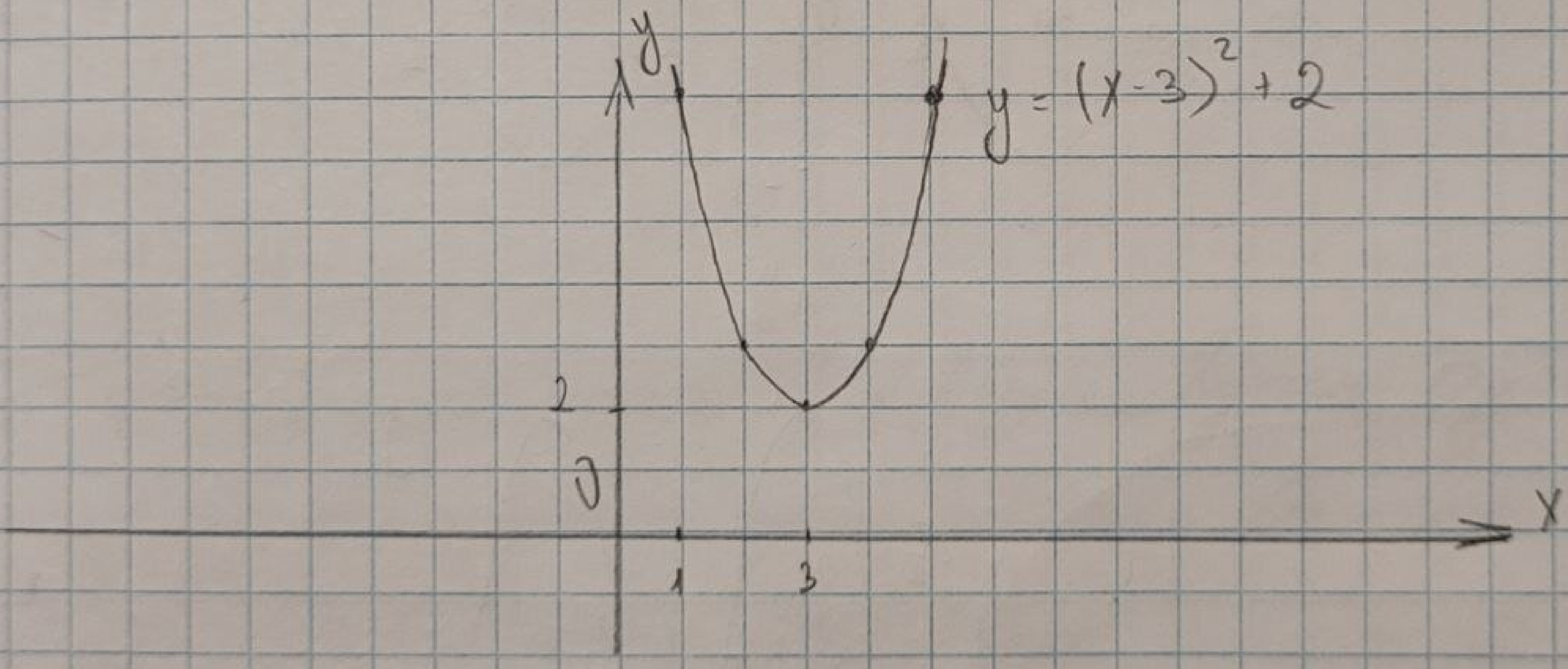
$\cancel{y}$ $x^2 - 6x + 11 = x^2 - 6x + 9 - 9 + 11 =$

$= (x-3)^2 + 11$

- $y = x^2$
- $y = (x-3)^2$, вправо на 3 ед
- $y = (x-3)^2 + 2$, вверх на 2 ед.


y
$y = x^2$
4
1
-1
-2
0
1
2
-1
x

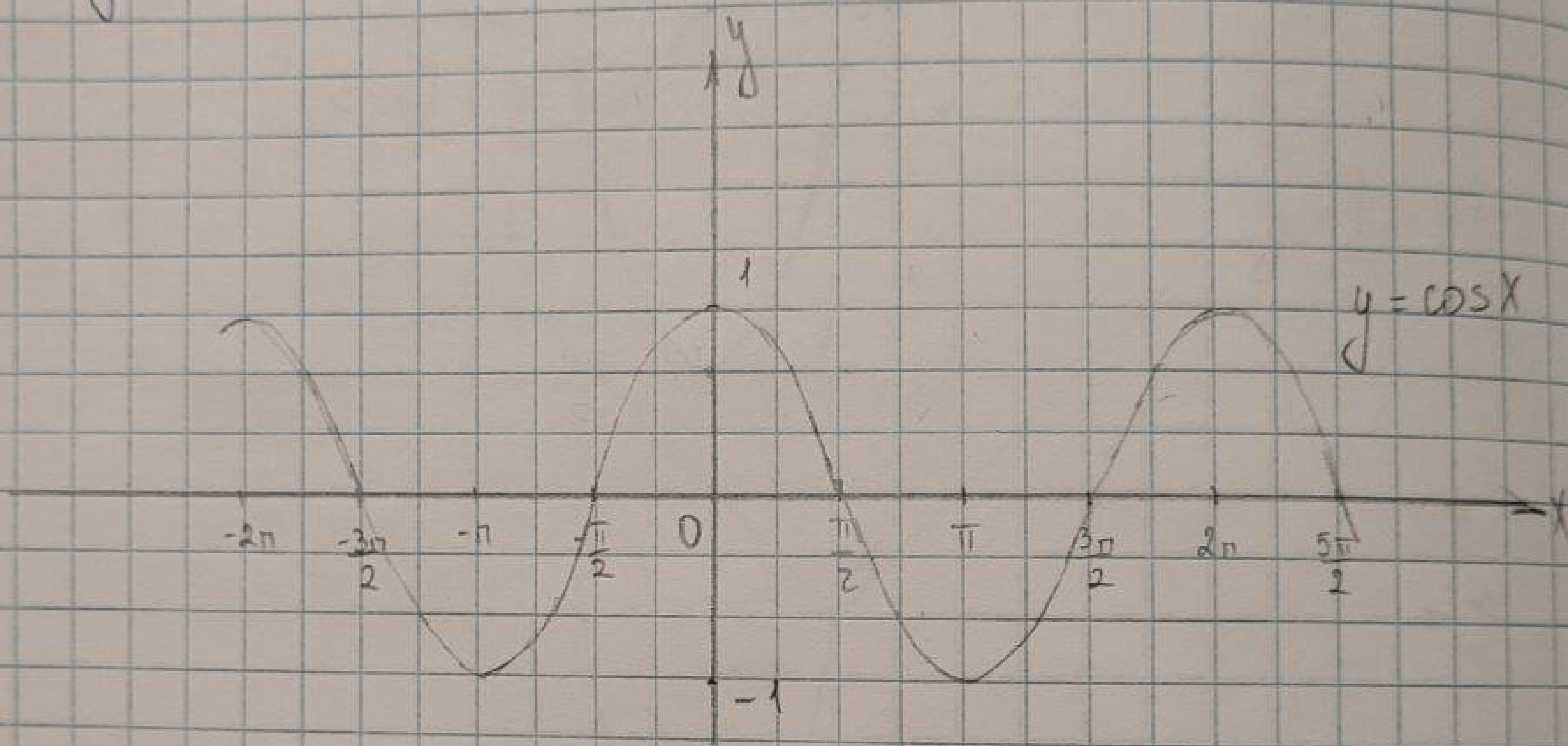

y
$y = (x-3)^2$
4
1
-1
1
2
3
0
4
5
x

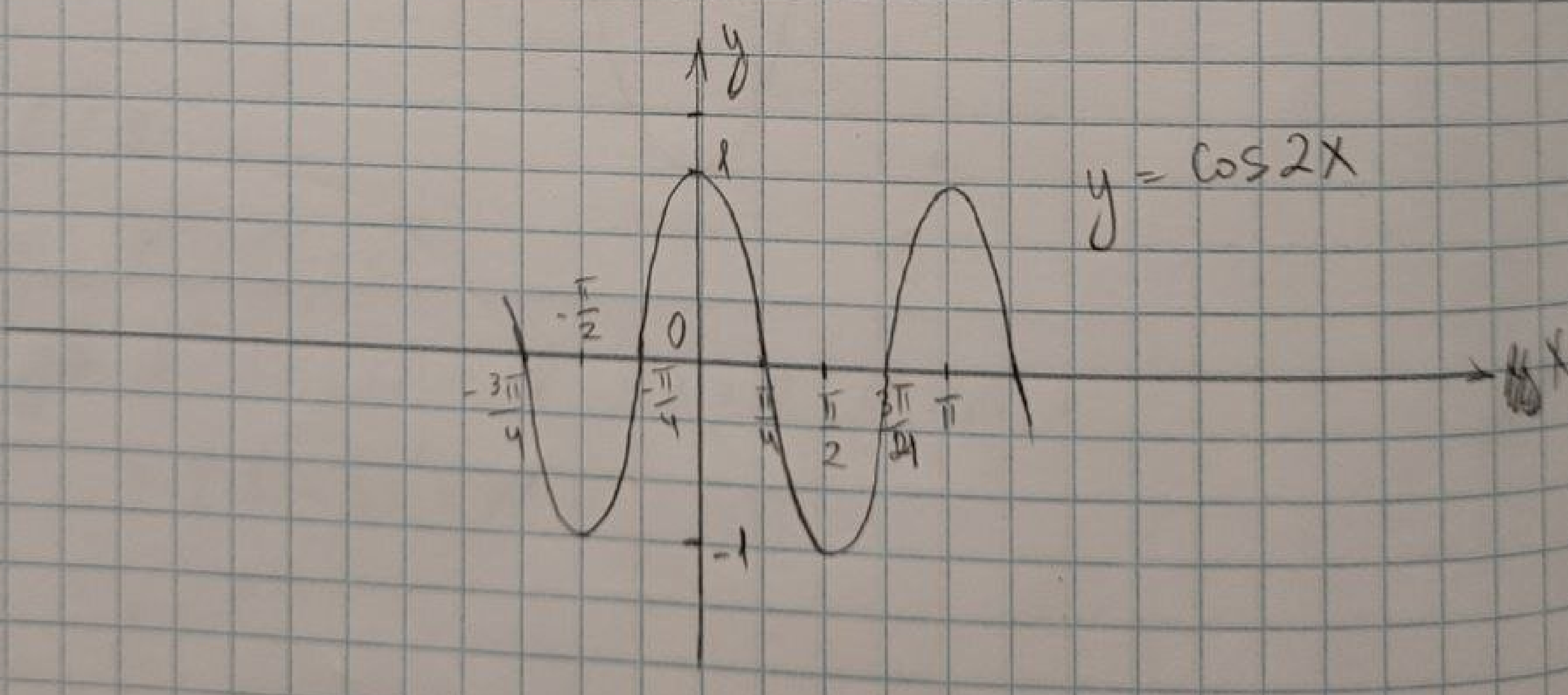

y
$y = (x-3)^2 + 2$
2
0
1
3
x

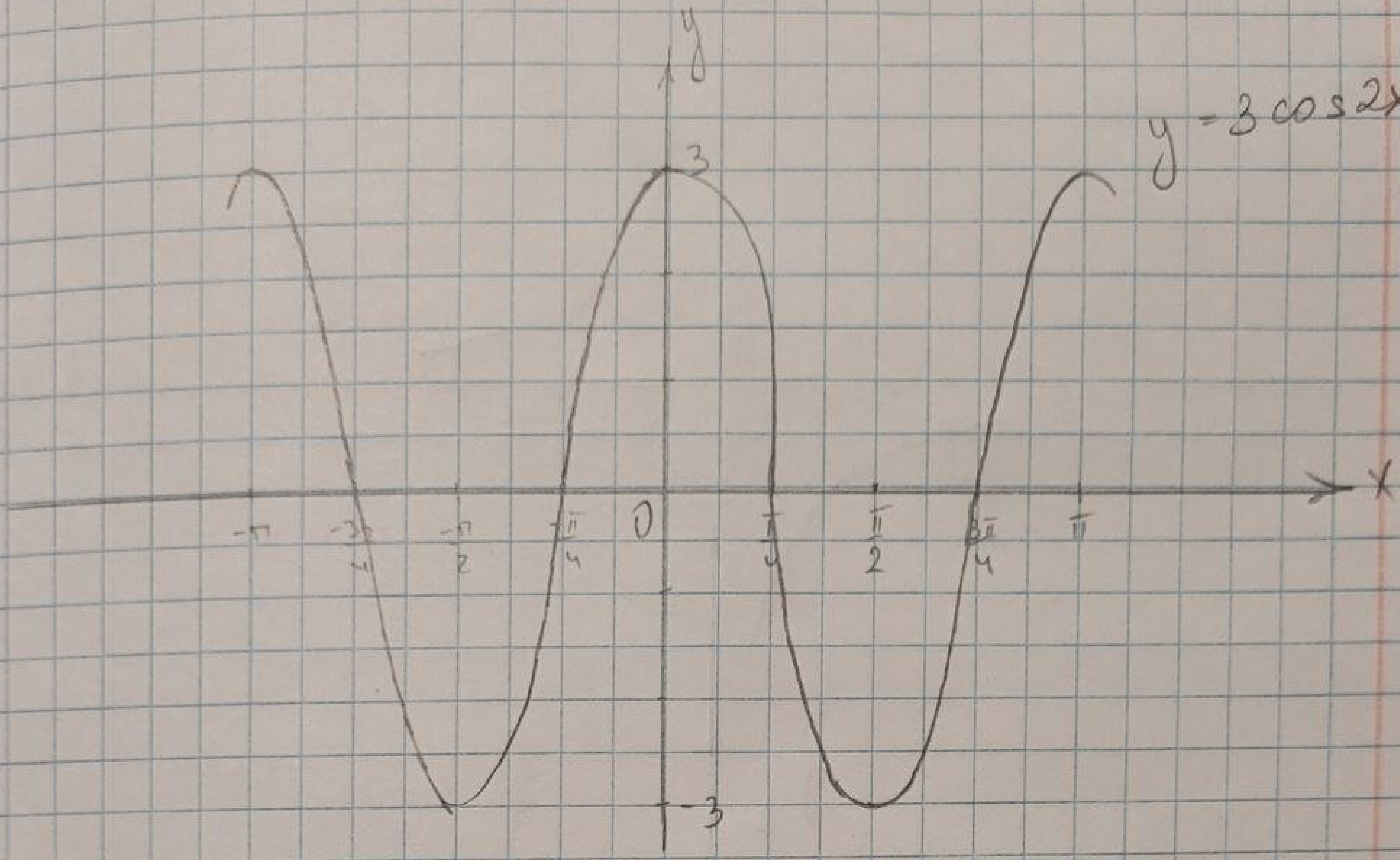
6.1.31

$y = 3\cos 2x$

- $y = \cos x$
- $y = \cos 2x$, сжатие в 2 раза вдоль Ox
- $y = 3\cos 2x$. растяж. в 3 раза вдоль Oy



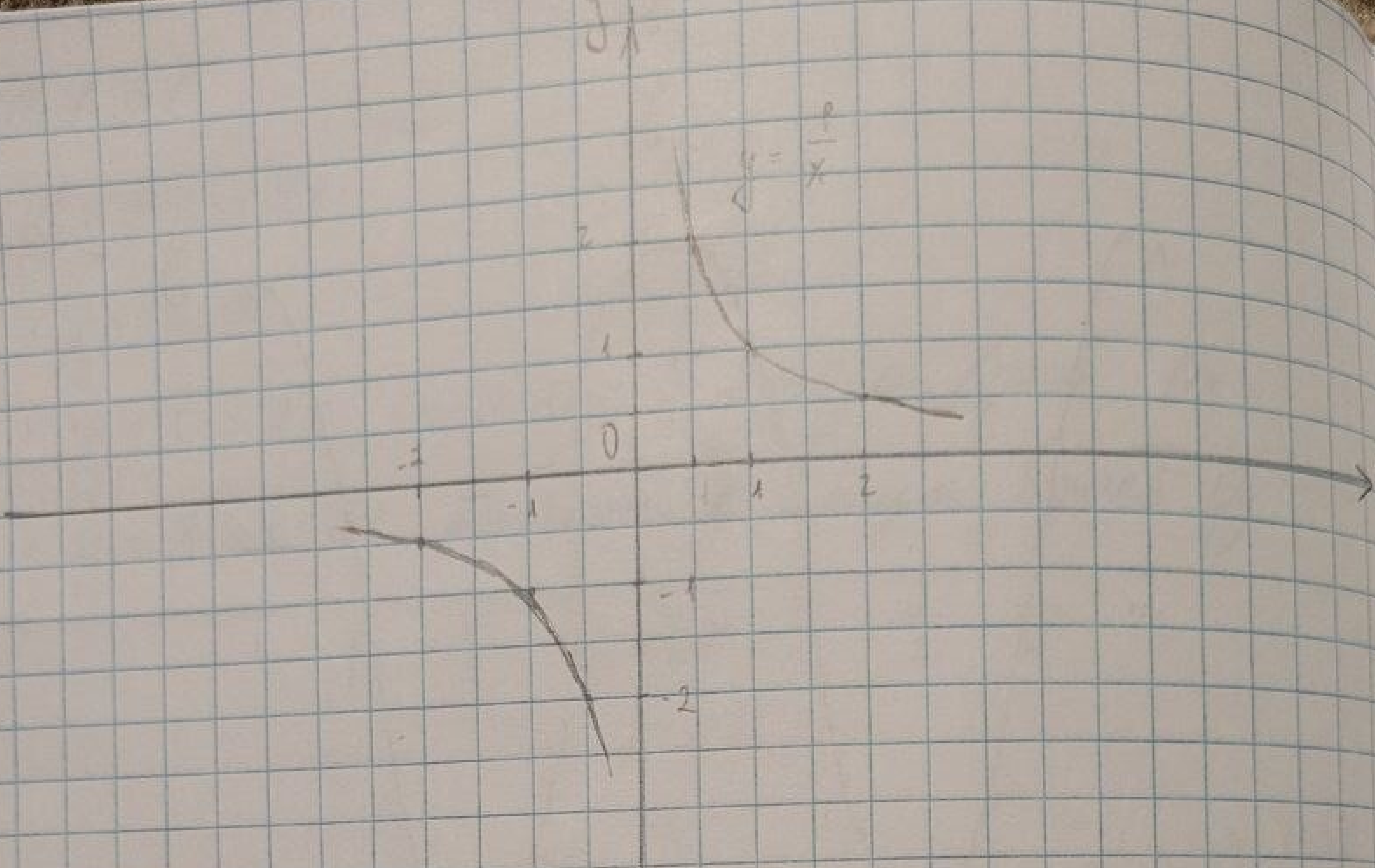




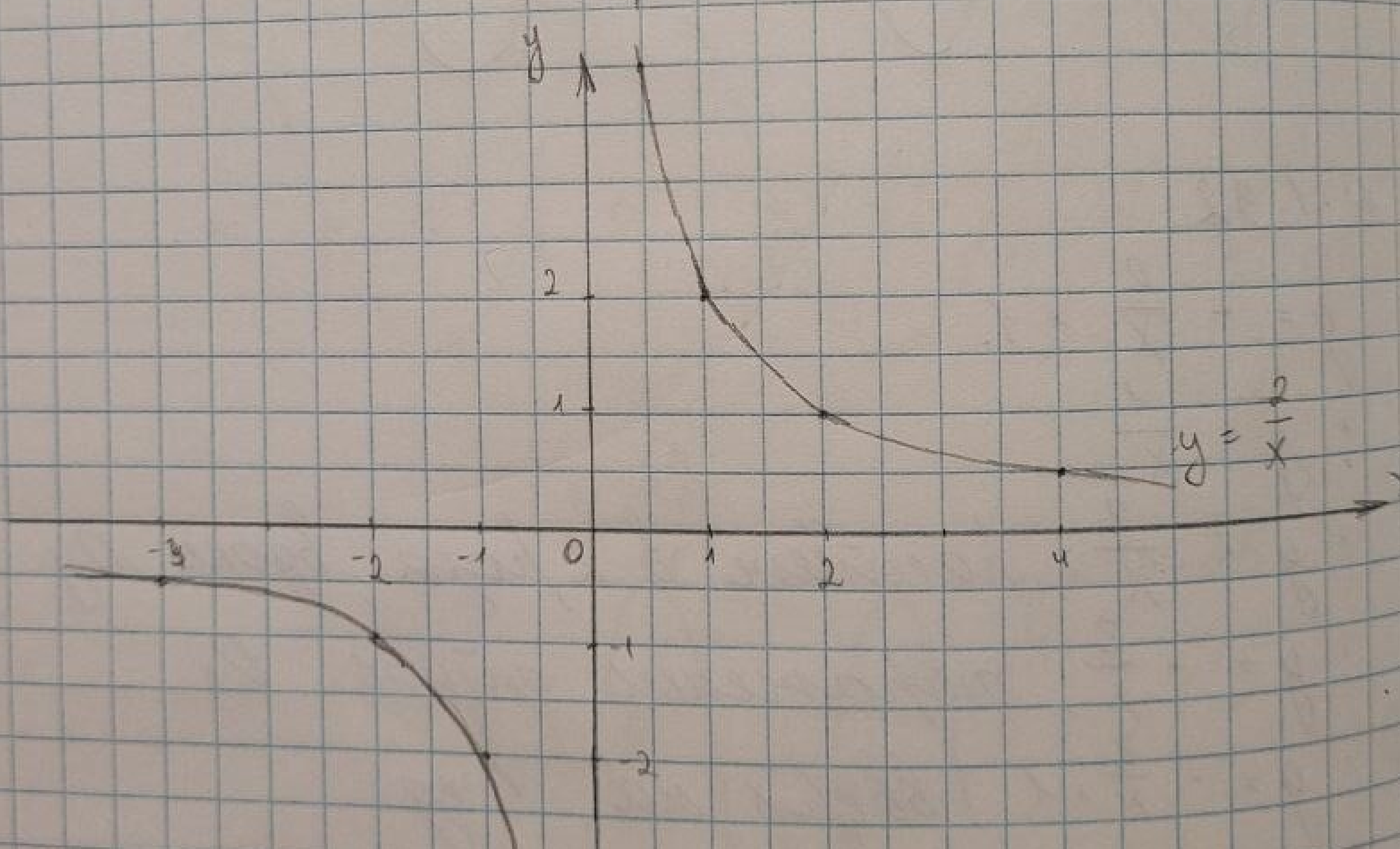
б. 1.32

$y = -\frac{2}{x} + 1$

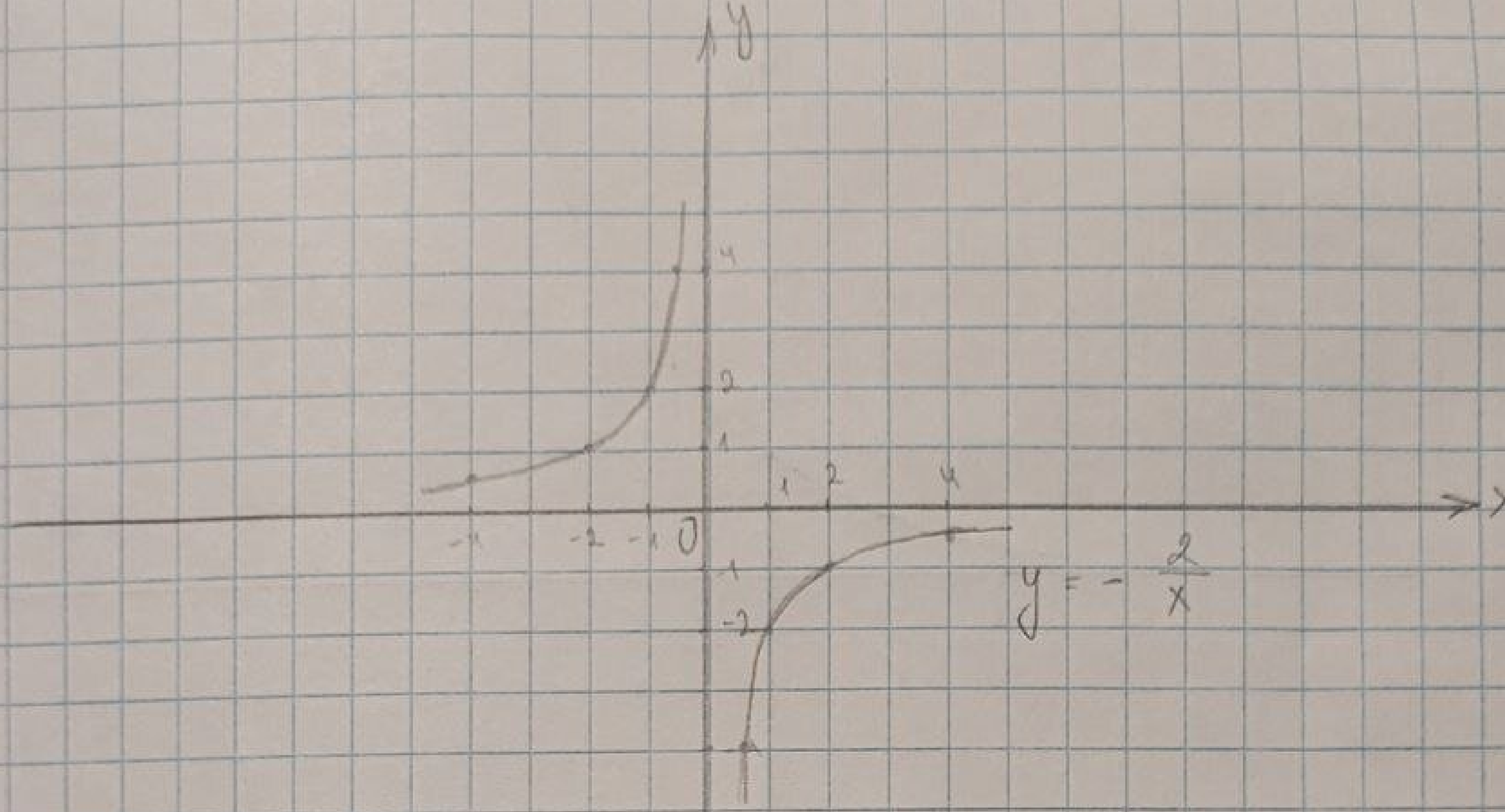
- $y = \frac{1}{x}$
- $y = \frac{2}{x}$, растяжение в 2 раза вдоль Oy
- $y = -\frac{2}{x}$, отражение относ. Ox
- $y = -\frac{2}{x} + 1$, вверх на 1 ед

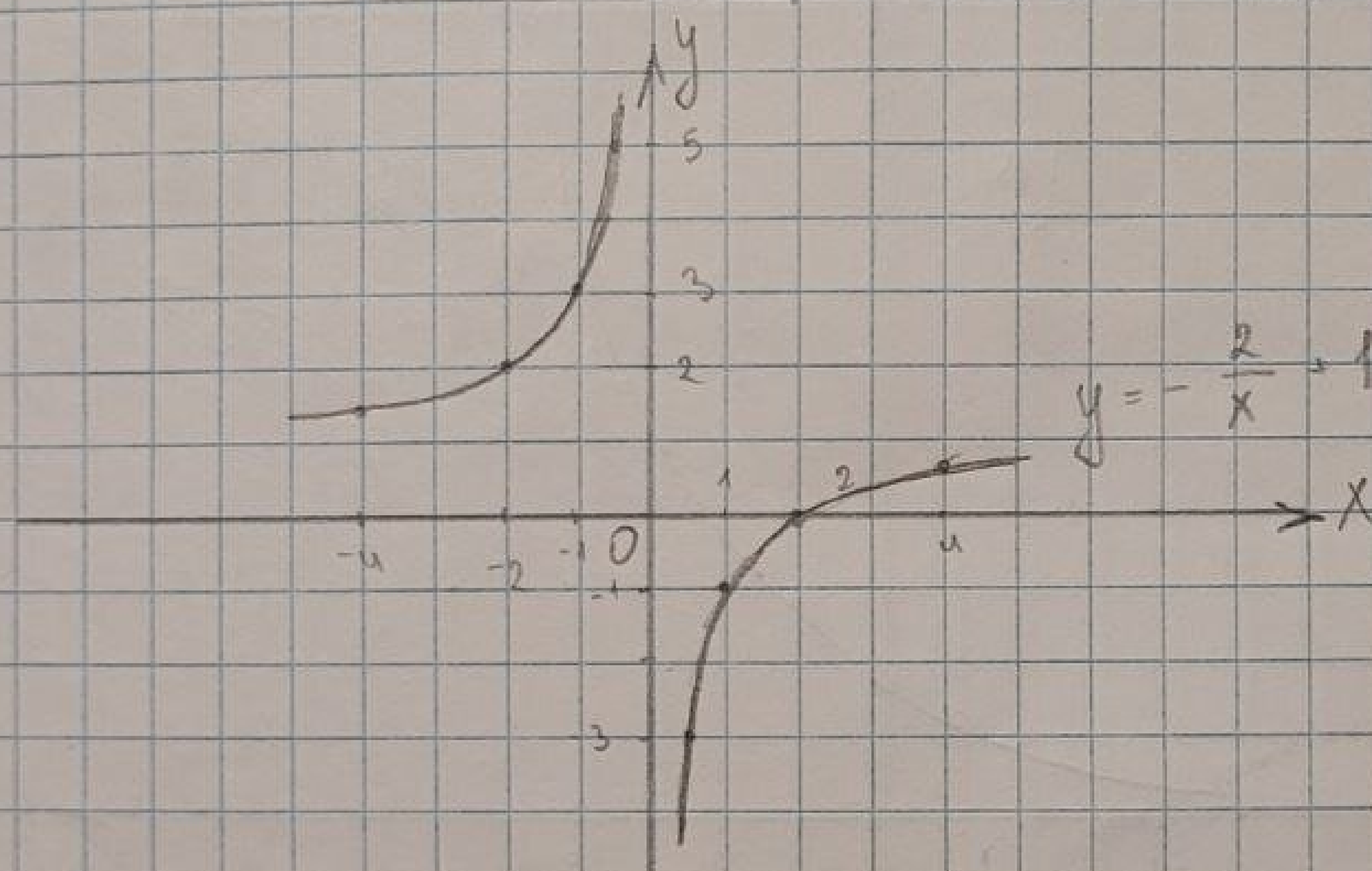


y
$y = \frac{1}{x}$
2
1
0
-2
-1
1
2
x
-1
-2



y
2
1
$y = \frac{2}{x}$
x
-3
-2
-1
0
1
2
4
-1
-2
-4

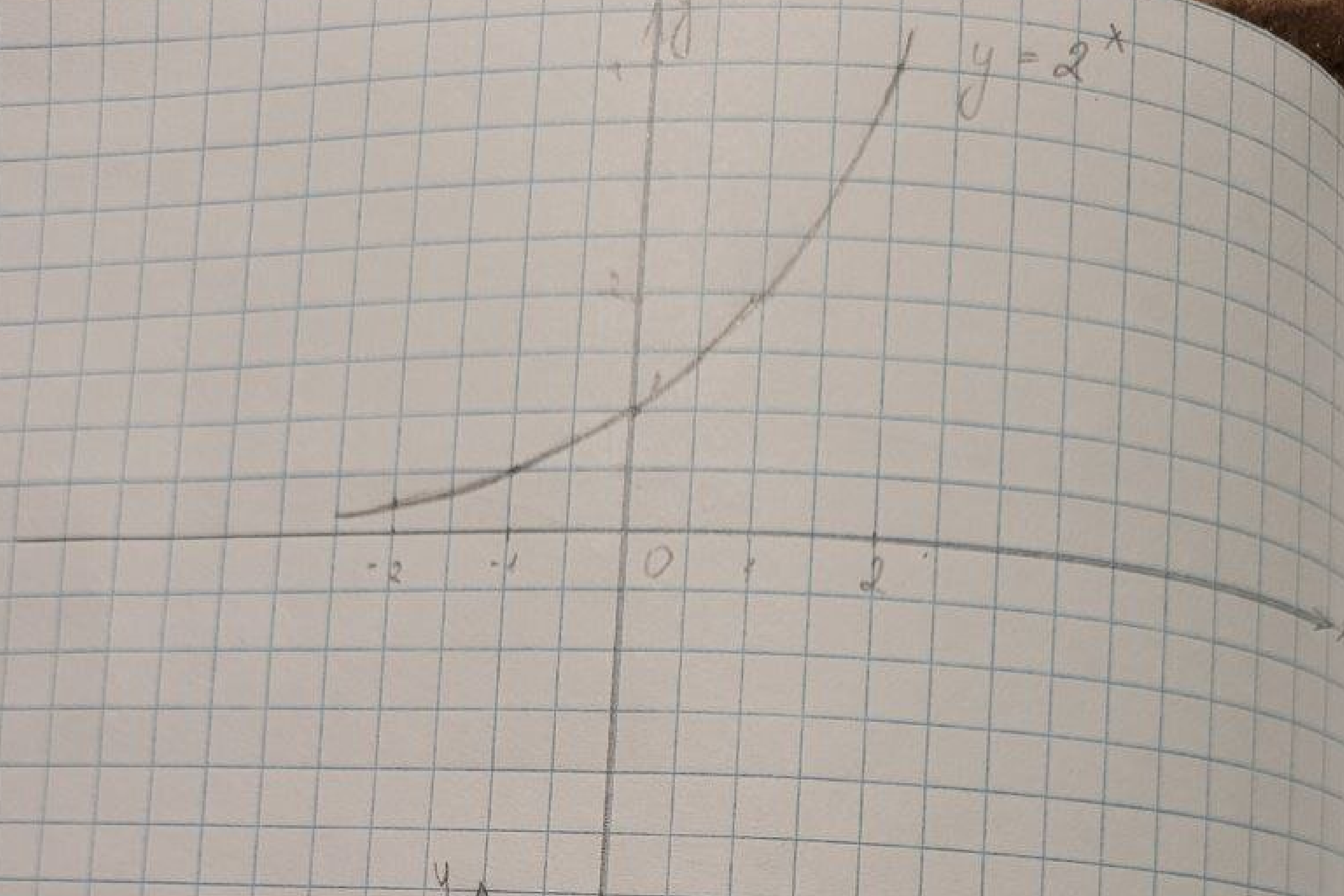




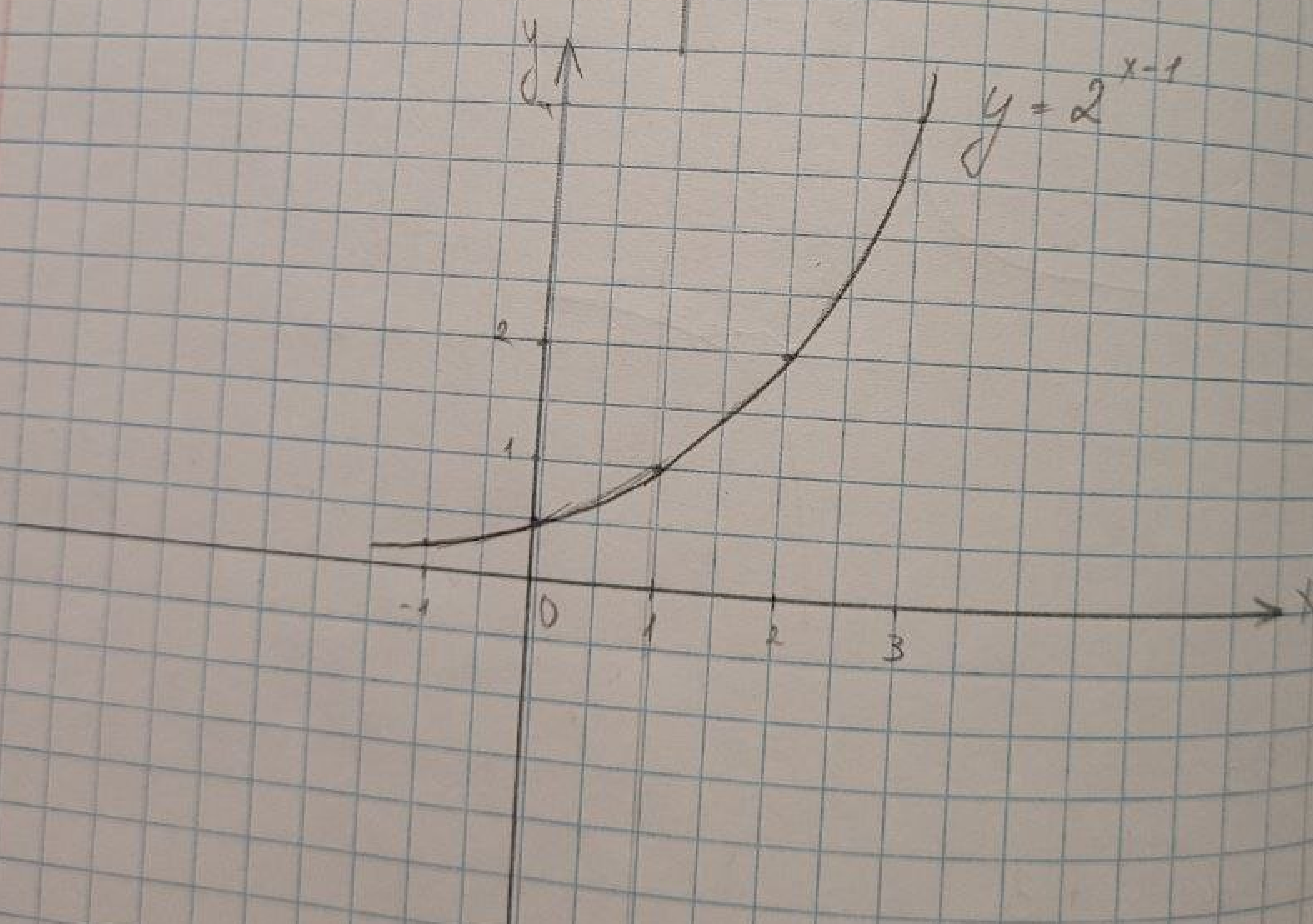
6.1.33

$y = 2^{x-1} + 3$

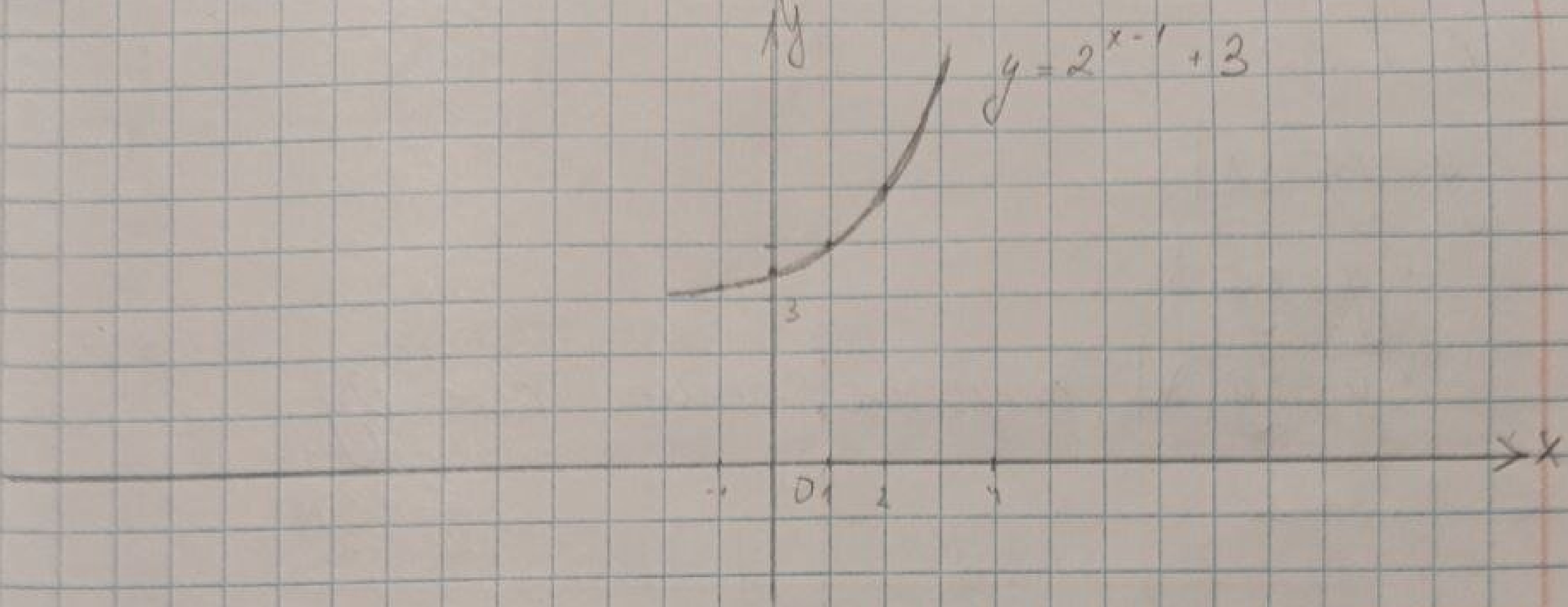
- $y = 2^{x}$
- $y = 2^{x-1}$, вправо на 1 ед
- $y = 2^{x-1} + 3$, вверх на 3 ед.



y
y = 2^x
-2
-1
0
1
2



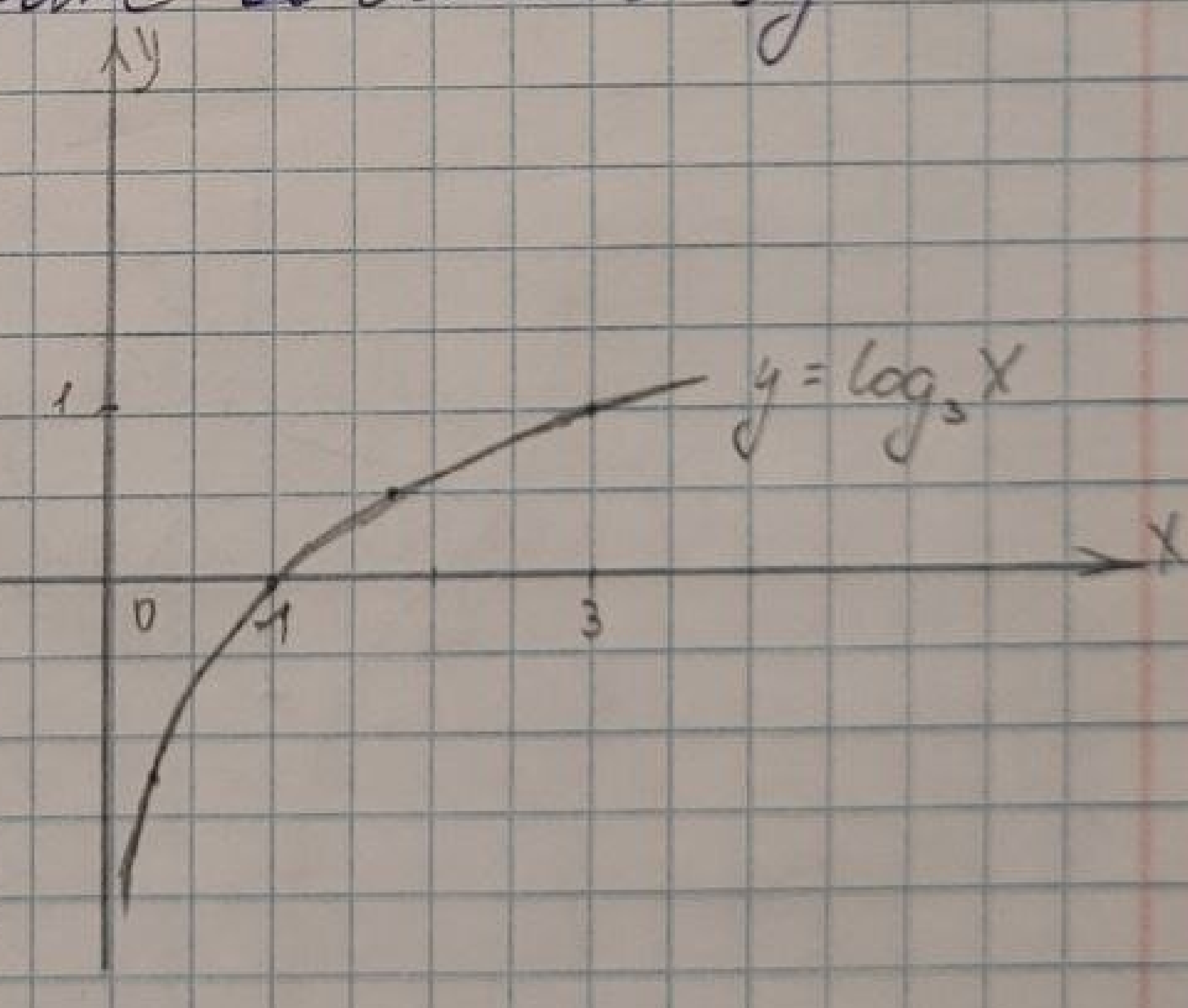
y
y = 2^{x-1}
x
2
1
-1
0
1
2
3

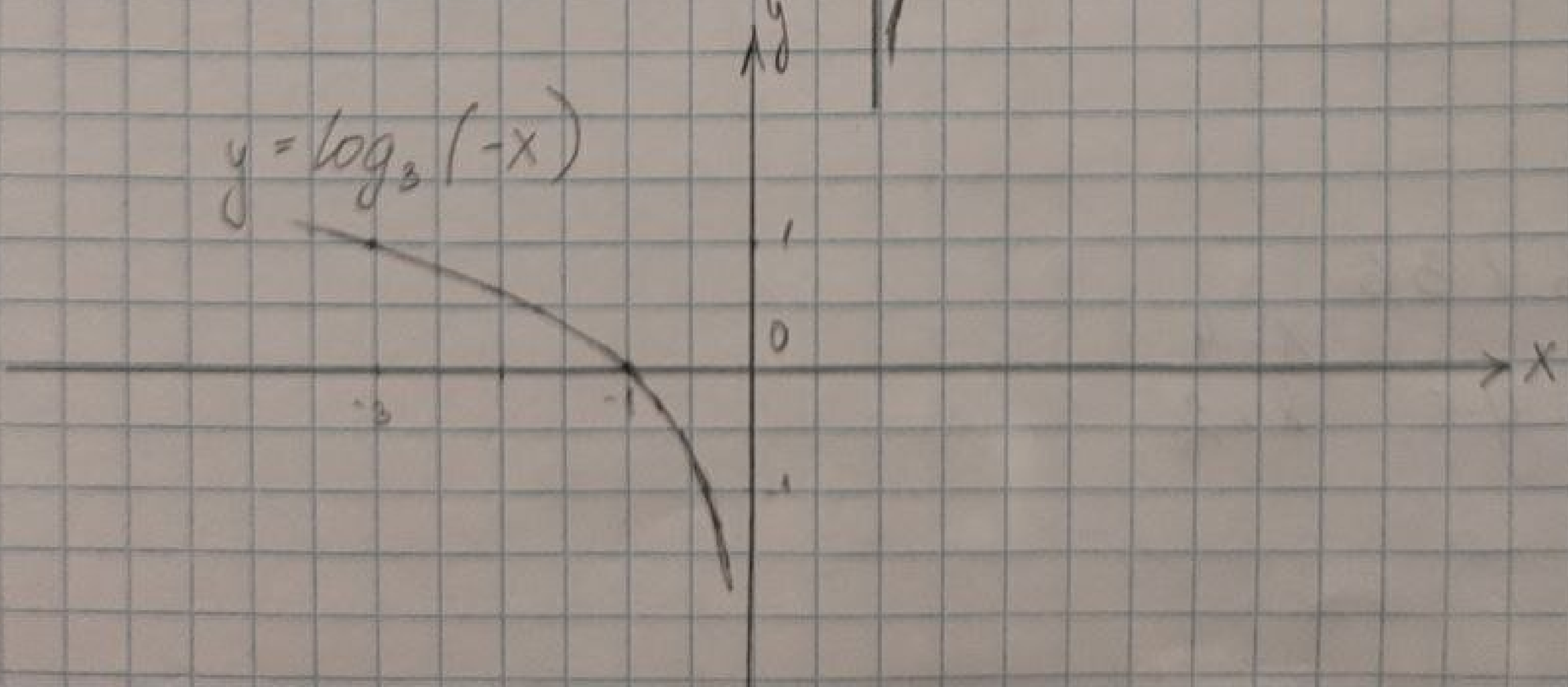


6.1.34

$y = \log_3(-x)$

- $y = \log_3(x)$
- $y = \log_3(-x)$, отраж. относ Oy



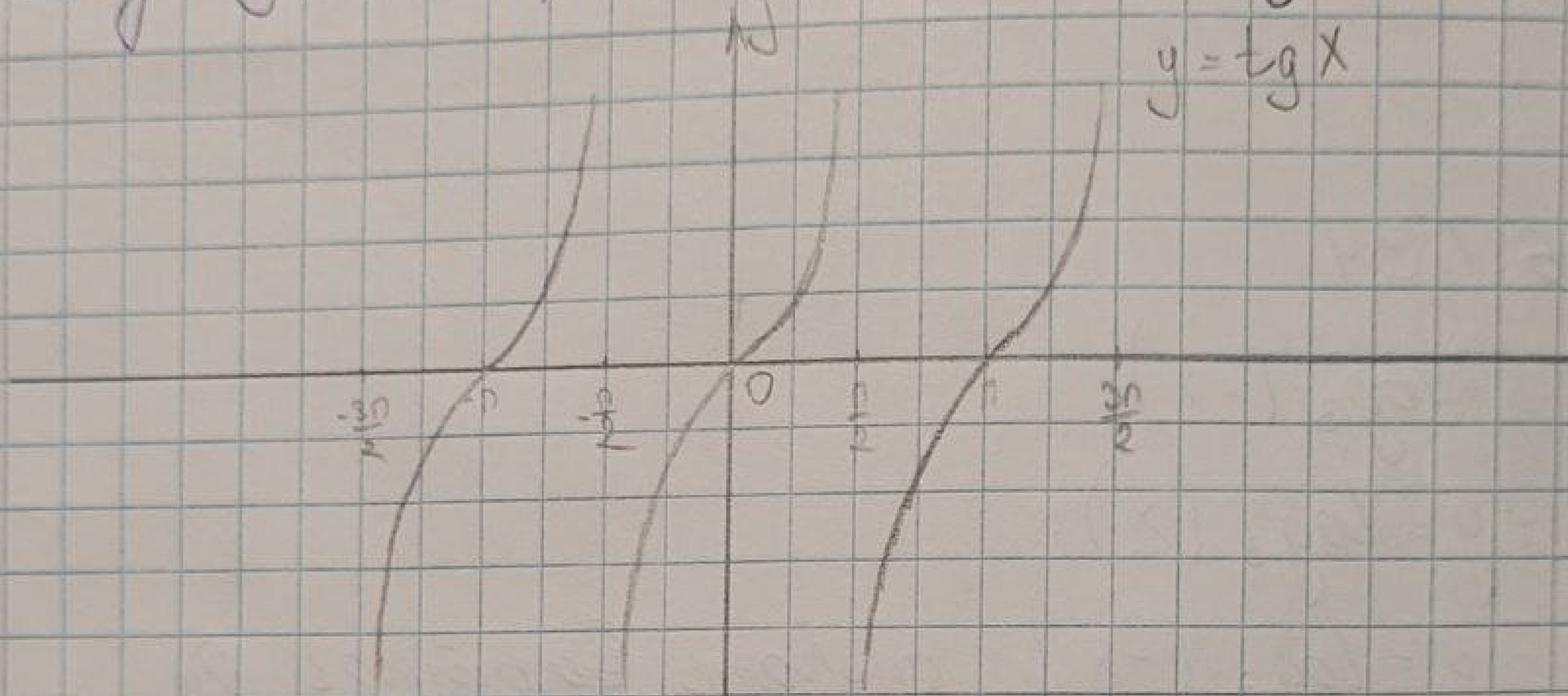


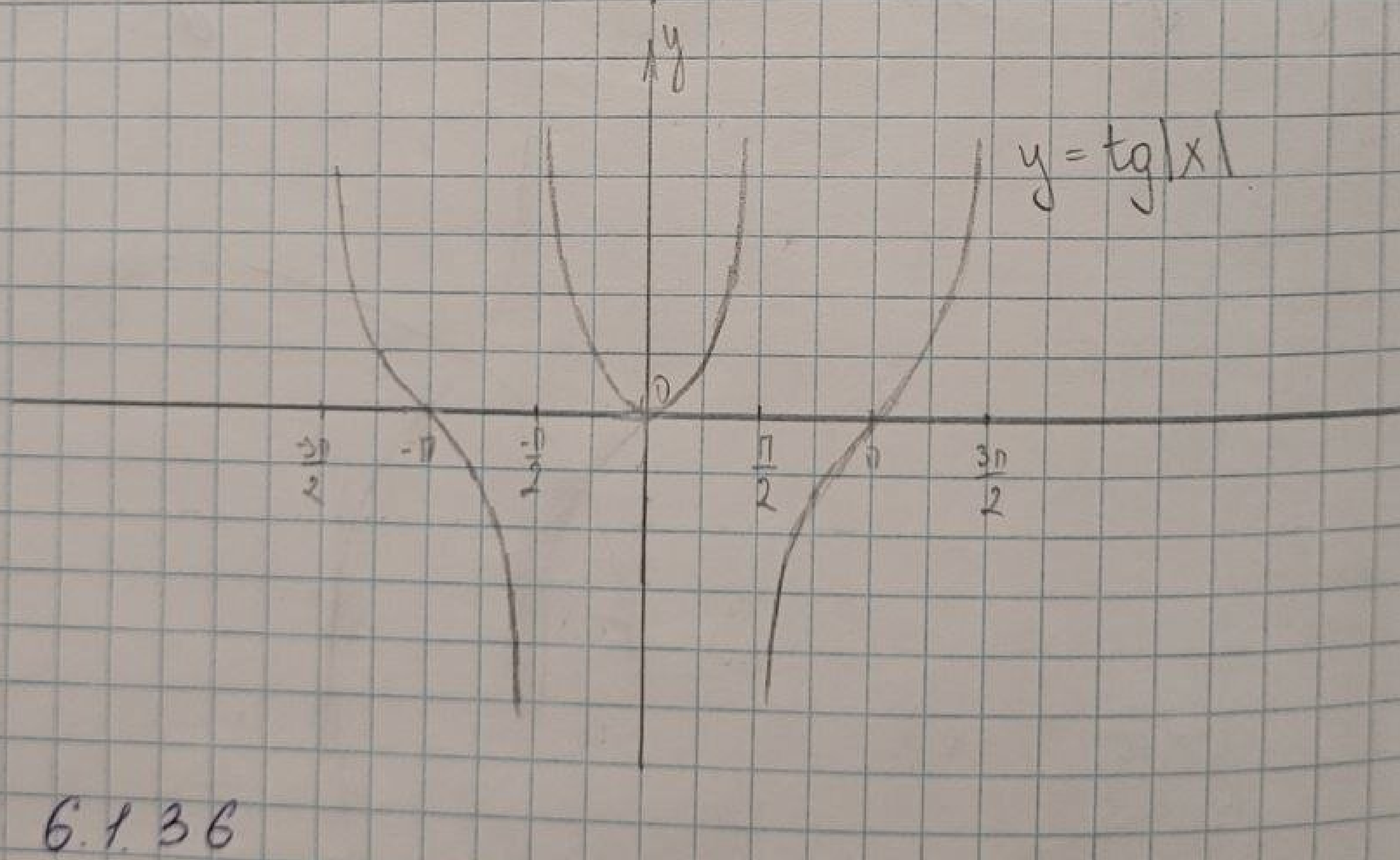
6.1.35

$y = tg|x|$

· $y = tg\,x$

· $y = tg|x|$, отраж. относ. оси Oy



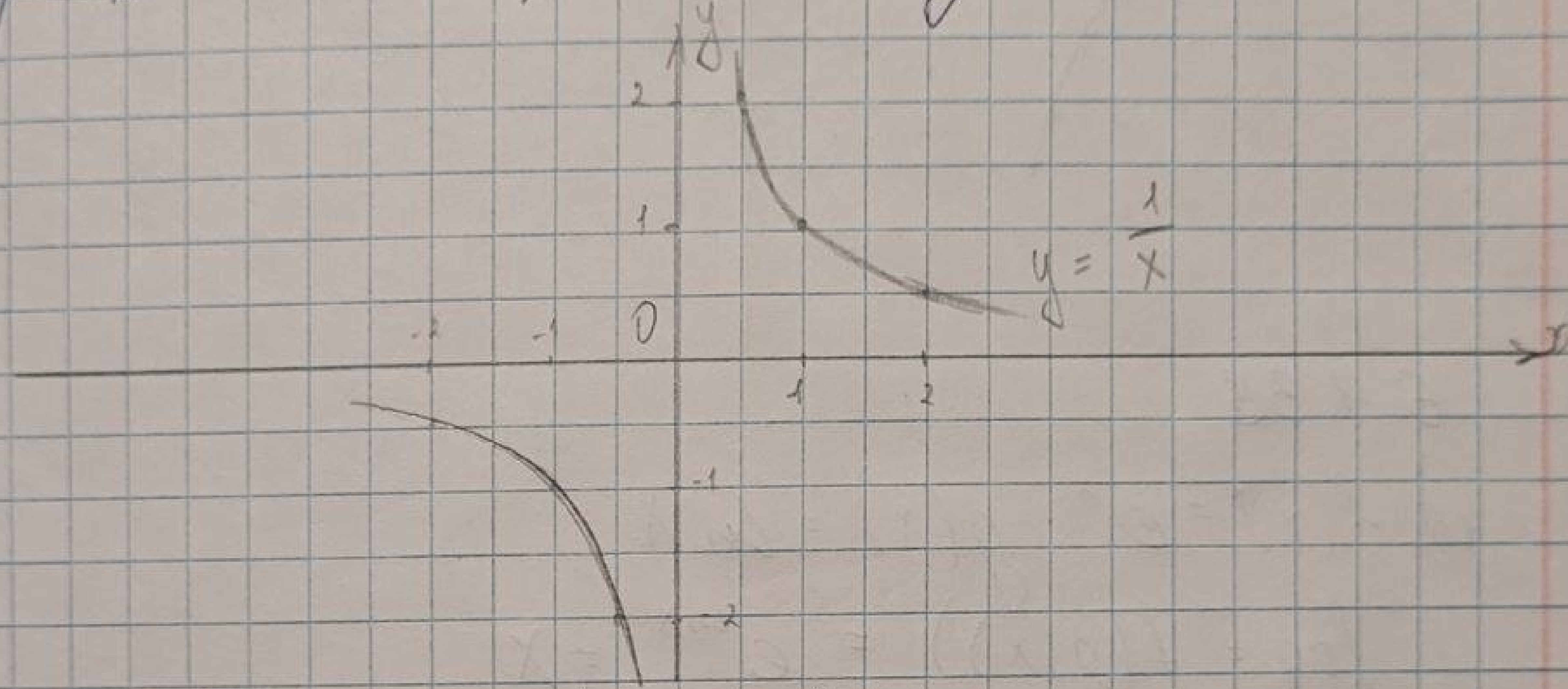


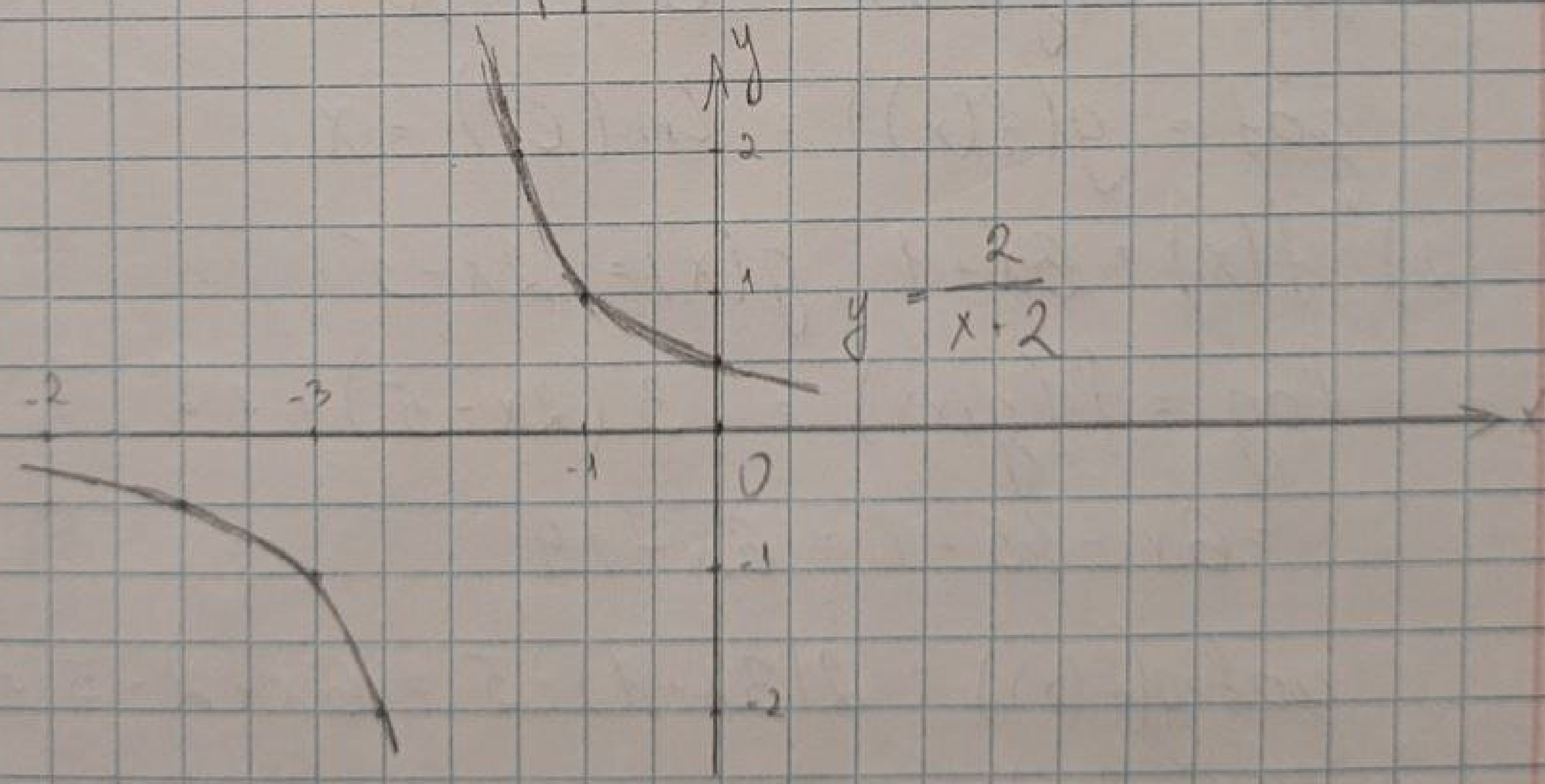
6.1.36

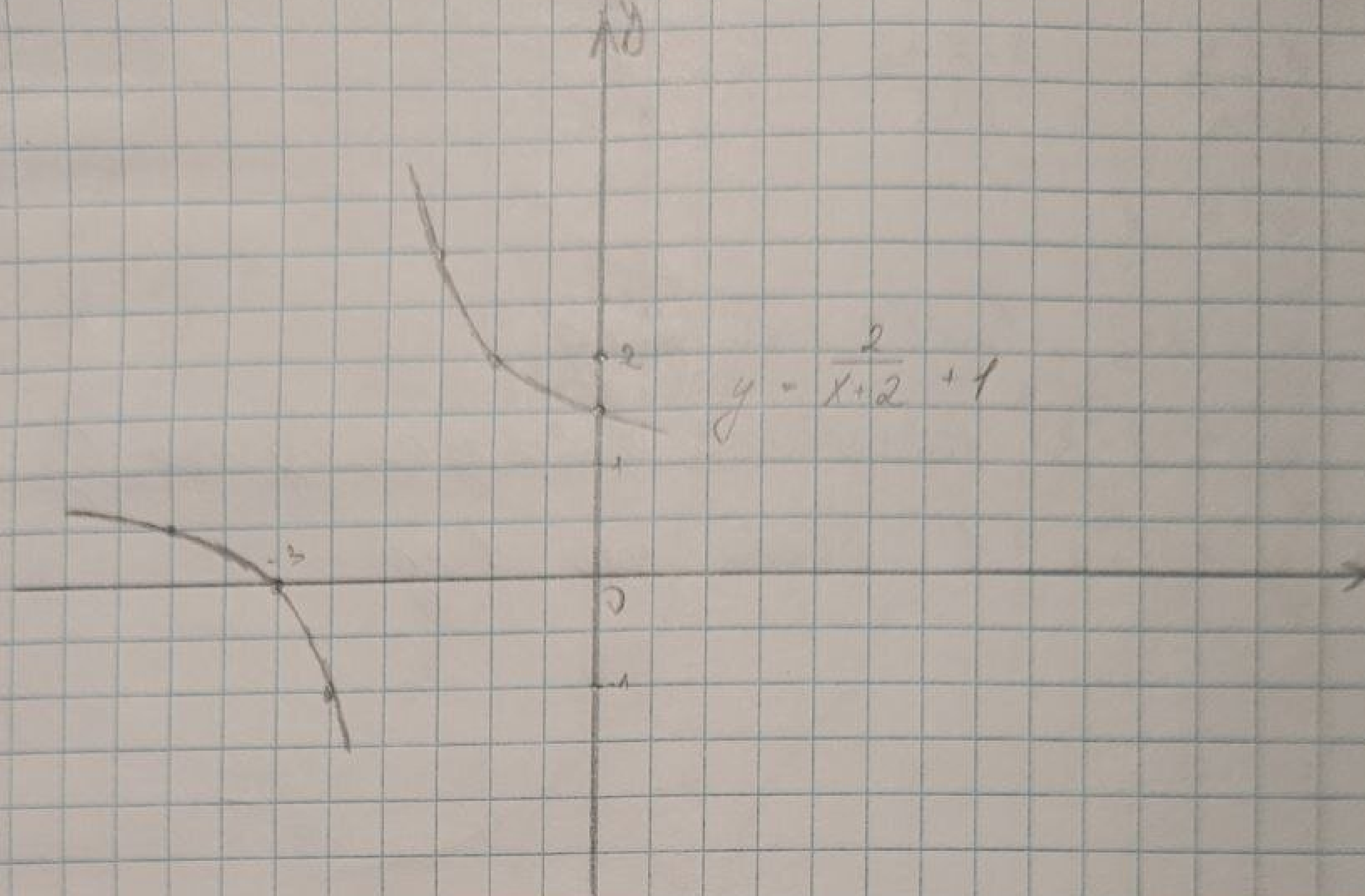
$$y = \frac{x+4}{x+2}$$

$$\frac{x+4}{x+2} = \frac{x+2+2}{x+2} = 1 + \frac{2}{x+2}$$

- $y = \frac{1}{x}$
- $y = \frac{1}{x+2}$, влево на 2 ед
- $y = \frac{2}{x+2}$, растяж в 2 раза вдоль Oy
- $y = \frac{2}{x+2} + 1$, вверх на 1 ед







6.1.38

1) $f(x) = e^x \quad g(x) = \ln x$

$f \circ g = f(g(x)) = e^{\ln x} = x$

$g \circ f = g(f(x)) = \ln(e^x) = x$

2) $f(x) = 3x + 1 \quad g(x) = 2x - 5$

$f \circ g = f(g(x)) = 3(2x-5) + 1 =$

$= 6x - 15 + 1 = 6x - 14$

$g \circ f = g(f(x)) = 2(3x+1) - 5 = 6x + 2 - 5 = 6x - 3$

3) $f(x) = |x| \quad g(x) = \cos x$

$f \circ g = f(g(x)) = |\cos x|$

$g \circ f = g(f(x)) = \cos|x|$

6.1.41

$y = 3x + 5$

$\square$ · на $(-\infty; \infty)$ $f(x) \uparrow$, т.е $x_1 \neq x_2 \Rightarrow$

$f(x_1) \neq f(x_2) \Rightarrow \exists$ обрат ф-ия

· $x = \frac{y-5}{3} \Rightarrow y^{-1} = \frac{x-5}{3}$ $\blacksquare$

6.1.42

$y = x^3 - 2$

$\square$ · на $(-\infty; \infty)$ $f(x) \uparrow$, т.е $x_1 \neq x_2 \Rightarrow$

$f(x_1) \neq f(x_2) \Rightarrow \exists$ обрат ф-ия

· $x = \sqrt[3]{y+2} \Rightarrow y^{-1} = \sqrt[3]{x+2}$ $\blacksquare$

6.1.43

$y = |x|$

$\square$ на $(-\infty; 0]$ $f(x) \downarrow$; на $(0; \infty) \uparrow$

$\forall y_0 > 0 \quad y_0 = |x| \Rightarrow \begin{matrix} x_1 = -x \\ x_2 = x \end{matrix} \Rightarrow$

· $\Rightarrow \nexists\, y^{-1}$, т.к $\cap$ граф. в 2ух точках

$\blacksquare$

6.1.43

$y = \frac{x-2}{x}$

$\square$ $D(f): x \in \mathbb{R} \setminus \{0\}$

на $(-\infty; \infty)$ $f(x)\uparrow$, т.е. $x_1 \neq x_2 \Rightarrow$

$\Rightarrow f(x_1) \neq f(x_2) \Rightarrow \exists\, y^{-1}$

$x = \frac{y-2}{y}$; $xy = y-2$; $y - xy = 2$; $y(1-x) = 2$

$\Rightarrow y^{-1} = \frac{2}{1-x}$

$\blacksquare$

6.1.45

$f(x) = c$

$\square$ Ф-ия монотона ($f(x)$ не $\uparrow$ и не $\downarrow$)

Ф-ия ограничена

$\blacksquare$

6.1.46

$f(x) = \sin^2 x$

$\square$ Огранич. ф-ия

$[-1; 1]$

$\blacksquare$

6.1.47

$f(x)=\operatorname{arctg} x$

□ Строго монотонная ($f(x)$ ↑)

Ограничен.

$\left[-\frac{\pi}{2}; \frac{\pi}{2}\right)$

51

53

56

■

6.1.48

$f(x)=-x^2+2x$

□ не явл. ни монотонной, ни строго монот., ни огранич.

на $(-\infty; 1)$ $f(x)$ ↑ на $[1, \infty)$ ↓

■

6.1.49

$f(x)=\frac{x+2}{x+5}$

□ Строго монотонная ф-ия

на $(-\infty; \infty)$ $f(x)$ ↑

■

6.1.54

□ 1) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

$\frac{y^2}{4} = \frac{x^2}{9} - 1 \quad | \cdot 4$

$y^2 = \frac{4x^2}{9} - 4$

$y = \pm\sqrt{\frac{4x^2}{9} - 4}$

$y = \pm\sqrt{\frac{4x^2 - 36}{9}}$

$y = \pm 2/3\sqrt{x^2 - 9}$

2) $x + |y| = 1$

$|y| = 1 - x$

$y = \pm(1 - x)$ при $x \leq 1$

3) $e^y - \sin y = x^2$

$\boxed{\text{///}}$

6.1.55

$y + \cos y - x = 0$

□ 1) $A(1; 0)$

$0 + \cos 0 - 1 = 0 \quad 0 = 0 \Rightarrow$ принадл

3

4

2) $B(0;0)$

$0 + \cos 0 - 0 = 1 \quad 1 \neq 0 \Rightarrow$ нет

3) $C\left(\frac{\pi}{2}; \frac{\pi}{2}\right)$

$\frac{\pi}{2} + \cos\frac{\pi}{2} - \frac{\pi}{2} = 0 \quad 0 = 0 \Rightarrow$ принад

4) $D(\pi - 1; \pi)$

$\pi + \cos\pi - \pi + 1 = 0 \quad 0 = 0 \Rightarrow$ принад

■